

UNIT 3

Environmental Pollution – I

"When the quality of life goes down for environment, the quality of life goes down for humans".

— George Holland

Objectives

- Introduction
- Environment
- Atmosphere-Composition and Structure Composition
- Pollution
- Air Pollution
- Some Common Air Pollutants
- Environmental Effects of Air Pollution
- Air Pollution Control
- Carbon Dioxide Sequestration
- Water Pollution
- Classification of Water Pollutants
- Waste water Treatment
- Waste Water Treatment Processes
- Waste Waters from Some Typical Industries

3.1 INTRODUCTION

The environmental pollution is one of the most challenging problem today. In spite of continuous efforts of our environmental engineers, scientists and an conscious society, our problem of environmental pollution remains the same. It may be a case of polluted water at some industrial discharge point, the chloroflouro carbons (CFCs) that are attacking the earth's ozone layer, the carcinogens that are contaminating our ground water or the carbon dioxide emissions leading to climate change. The endless list of environmental pollution is a cause of great concern. Moreover radiation hazards and hazardous waste management join the array of environmental pollution problems.

Human beings depend upon the quality of environment they are living in and they also exert a tremendous impact on the environment through:

- extraction of resources
- modification and manipulation of the environment
- adding to the pollution through the process of industrialisation.

All the three basic amenities of human existence viz. air, water and land are experiencing the serious ecological imbalance. *Water* is polluted by pollutants such as sewage, solid wastes and industrial effluents. *Air* is polluted by automobile exhausts, exhaust gases from industries. *Land* is polluted by hazardous waste disposal. These environmental problems are

becoming a serious threat to our existence. Furthermore, the effects of these problems seem to be far more longer than we anticipate. For example, CFC's emitted today will be around for *decades*, carbon dioxide from our power plants will be in the atmosphere for *centuries*, and our radioactive wastes will be problematic for *thousands of years*.

Hence there is a serious need to address these issues for harmonious existence of industry and the world at large.

Environmental science is defined as - "Applying knowledge from many disciplines to the study and management of the environment. It deals with the analysis of the conditions, circumstances and influences affecting life and how life in turn responds".

Before proceeding to the environmental pollution let us have a brief idea of our environment.

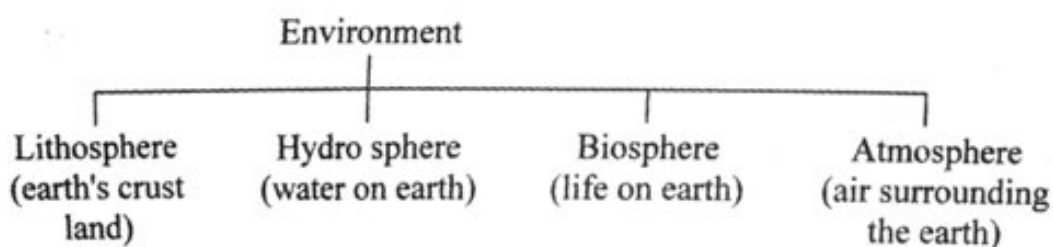
3.2 ENVIRONMENT

The human environment encompasses all physical, chemical, biological and social processes which exert a significant influence, directly or indirectly on human beings. In turn, human beings also exert influence on quality of environment. Thus, the subject of environment sciences has its roots in ecology which is the study of how living and non-living things in nature relate to one another.

The environment is not a simple distribution of gases but is highly complex and structured. It comprises of following four segments:

1. Lithosphere
2. Hydrosphere
3. Atmosphere
4. Biosphere

The names of the four sphere are derived from the Greek words for stone (litho), air (atom), water (hydro) and life (bio.).



Lithosphere

The lithosphere contains all of the hard solid land of the planet's crust (surface) and the semi-solid land underneath the crust. The surface of the lithosphere is very uneven. There are high mountain ranges, huge plains or flat areas and deep valleys along the ocean flow.

The solid and semi-solid land of the lithosphere form layers that are physically and chemically different. The outermost layer of the lithosphere consists of loose soil rich in nutrients, oxygen, and silicon. Beneath that layer lies a very thin, solid crust of oxygen and silicon. Next is a thick, semi-solid mantle of oxygen, silicon, iron, and magnesium. Below that is a

liquid outer core of nickel and iron. At the center of earth is a solid inner core of nickel and iron.

Generally, the word "lithosphere" means only the cold, hard surface of earth, not the entire inside of the planet.

Hydrosphere

The water on earth's surface constitute hydrosphere. It ranges from 10 to 20 kilometers in thickness. The hydrosphere extends from earth's surface downwards, several kilometers into the lithosphere and upwards about 12 kilometers into the atmosphere.

A small portion of the water in the hydrosphere is fresh (non-salty). This water flows as precipitation from the atmosphere down to earth's surface, as rivers and streams along earth's surface, and as groundwater beneath earth's surface. Most of earth's fresh water, however, is frozen.

Ninety-seven percent of earth's water is salty. The salty water collects in deep valleys along earth's surface. These large collections of salty water are called oceans. On the other hand, water near the poles is very cold, while water near the equator is very warm. The differences in temperature cause water to change physical states. Extremely low temperatures like those found at the poles cause water to freeze into a solid such as a polar ice-cap, a glacier, or an iceberg. Extremely high temperatures, like those found at the equator cause water to evaporate into a gas.

Frozen water (glaciers, icecaps, and icebergs) in its own sphere is called the "cryosphere".

Biosphere

The biosphere is composed of all living organisms. This sphere includes all of the microorganisms, plants, and animals of Earth. In fact, biosphere involves the interaction of the living beings with other segments. Most of the planet's life is found from three meters below the ground to thirty meters above it and in the top 200 meters of the oceans and seas.

Atmosphere

The atmosphere is the body of air which surrounds the earth. It extends from less than 1 m below the earth's surface to more than 348 km above the earth's surface. The upper portion of the atmosphere protects the organisms of the biosphere from the sun's ultraviolet radiation. It also absorbs and emits heat. When air temperature in the lower portion of this sphere changes, weather occurs. As air in the lower atmosphere is heated or cooled, it moves around the planet. The result can be as simple as breeze or as complex as a tornado endangering the survival of life. There is a constant exchange of matter taking place between the atmosphere, biosphere and hydrosphere.

The atmospheric temperature, pressure and density vary considerably with altitude.

- The temperature varies from -100°C to $+1200^{\circ}\text{C}$.
- Atmospheric pressure varies from 1 atmosphere (at sea level) to 3×10^{-7} atmosphere (at 100 km above sea level)
- The atmospheric density at the surface of the earth is about 0.0013 g/m^3 which decreases sharply with increasing altitude.

It is interesting to note that all four spheres can be and often are present in the single location. For example a piece of soil will have mineral material from the lithosphere, there will be elements of the hydrosphere present as moisture within the soil, the biosphere as insects and plants, and even the atmosphere as pockets of air between soil pieces. A detailed description of the structure of atmosphere is given in the next section.

3.3 ATMOSPHERE-COMPOSITION AND STRUCTURE

COMPOSITION

The proportions of the various constituents of the atmosphere remain more or less constant upto a height of about 16 km from the surface of earth. This region is sometimes referred to as homosphere. Above this the composition starts showing variability with altitude. In this region, called the heterosphere, gravitational separation takes place as the lightest and fastest atoms and molecules move farther from the earth.

Finally in the upper heterosphere hydrogen and heliums are the most typical gases found. The atmospheric composition of clean, dry air, near the sea-level is given in Table 3.1.

Table 3.1: Atmospheric constituents at sea-level [%]

N_2	78.1
O_2	20.9
Ar	0.93
CO_2	0.035
Ne	0.0018
He	0.00052
CH_2	0.0002
Kr	0.00011
H_2	0.00005
Xe	0.00001

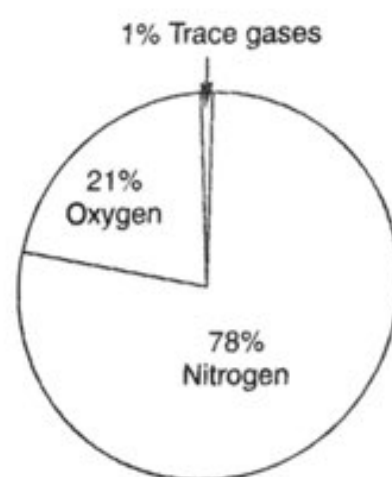


Fig. 3.1: Composition of Atmosphere

The Layers of the Atmosphere

The earth's atmosphere is highly structured. Because of earth's gravitational pull, the atmosphere is horizontally stratified. It can be divided into four zones depending on the temperature profile. (Fig. 3.2)

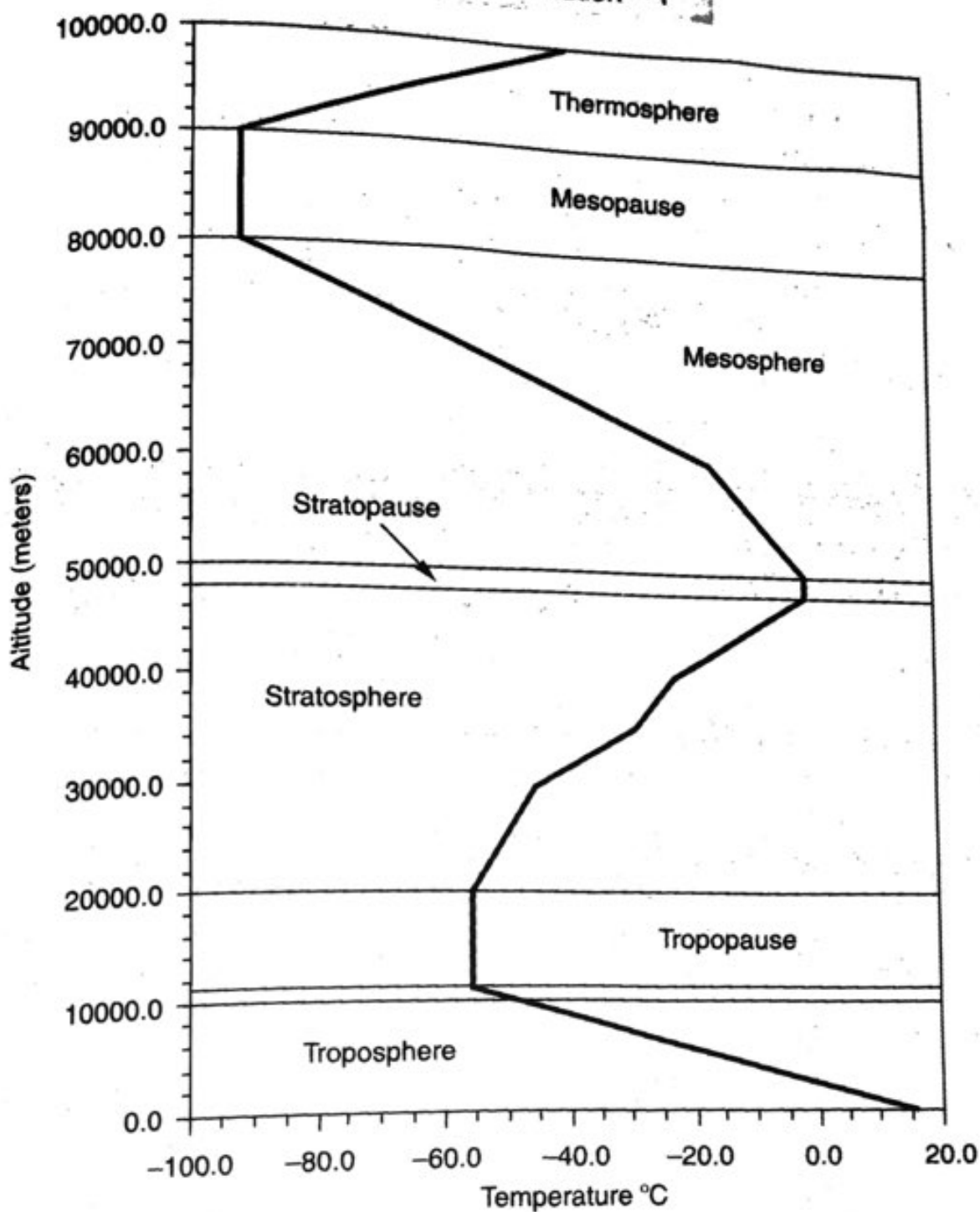


Fig. 3.2 Layers of atmosphere.

- (i) Troposphere
- (ii) Stratosphere
- (iii) Mesosphere
- (iv) Thermosphere

(i) **Troposphere:** The word troposphere comes from tropein, meaning to turn or change. All of the weather occurs in the troposphere.

It has the following characteristics:

- It is nearest to the earth's surface and extends from the earth's surface to an average of 12 km.
- The temperature generally decreases with increasing height upto the top of the troposphere (tropopause). It drops from about 17°C to - 52°C.
- The troposphere ends at a point where temperature no longer varies with height. This area is known as tropopause. Tropopause marks the transition to the stratosphere.

- The density of air, in this region, decrease exponentially with increasing altitude. The lower density of molecules higher up would not give us enough oxygen to survive.
- The moisture content also decreases with height upto tropopause.
- Towards the top of the troposphere speed of the winds increases.
- The tropopause and troposphere are known as lower atmosphere.
- The top of the troposphere is extremely cold and windy place.

(ii) **Stratosphere:** The gradual change from the troposphere to the stratosphere begins at approximately 11 km high. The stratosphere has following characteristics.

- It extends upto 50 km high.
- It is dry and less dense than troposphere.
- The temperature in the lower stratosphere is extremely stable and cold at -57°C . However, from the middle of the stratosphere, and up, the temperature pattern changes, sharply increasing with height. This is due to increasing levels of ozone concentration which absorbs ultra violet radiation. The temperature reaches to 18°C in the upper atmosphere at 40 km altitude.
- The ozone layer is in the stratosphere.
- The stratopause separates the stratosphere from the next layer.

(iii) **Mesosphere:** The mesosphere has the following characteristics:

- It starts just above the stratosphere and extends to 85 km high.
- The temperature again falls in this region. It reaches to as low as -93°C with increase in altitude. This is due to low levels of ozone and other species that can absorb ultra violet species from sun.
- Mesosphere is separated from thermosphere by mesopause. Mesopause is the region of coldest temperature in the atmosphere (i.e. about -100°C).
- The regions of the stratosphere and mesosphere along with stratopause and mesopause are called middle atmosphere.

(iv) **Thermosphere:** The transition from the mesosphere to the final thermosphere layer begins at a height of approximately 81 km. Thermosphere has the following characteristics:

- The temperature rises very rapidly with attitude and hence the name thermosphere. The maximum temperature can reach a staggering 1982°C . These extreme temperatures are caused by the absorption of the sun's shortwave ultraviolet radiation.
- The UV radiation penetrates the upper atmosphere and ionizes the atoms. The charged atoms build up to form a series of layers within the thermosphere. This charged layer is sometimes referred to ionosphere. The ionosphere can deflect some radio signals.

3.4 POLLUTION

The earth's environment has been composed in a very unique way for sustaining the life on earth. Through one process, the different constituents of the atmosphere are consumed and through the other, these constituents are released into the atmosphere.

In other words, the composition of atmosphere remains constant. The oxygen cycle in the air provides a very good example. During respiration, oxygen is consumed by living beings and is again released into the atmosphere by plants through a process called photosynthesis.

The things would have that simple, had the man not interfered with the nature. But in the era of industrial revolution, the composition of the atmosphere has been changed undesirably against the nature. Thus we have increased proportions of different harmful constituents in atmosphere and decreased amount of oxygen. In other words, we have polluted the atmosphere. The same is the case with water and soil.

Thus, pollution can be defined as any undesirable change in the composition of air, water and soil or any segment of environment against nature. The pollution results in deteriorating the nature and quality of the environment.

The undesirable or unwanted foreign species causing pollution in air, water and soil are called *pollutants*.

On the basis of factors causing pollution, pollution can be classified under the following heads:

1. Air pollution
2. Water pollution
3. Soil pollution
4. Solid waste pollution
5. Hazardous waste pollution

We will discuss each one of these under the following sections.

3.5 AIR POLLUTION

Air pollution can be defined as the presence of chemicals in the atmosphere in quantities and duration that are harmful to human health and the environment. It occurs when the concentration of certain substances become high enough to cause the atmospheric environment to become toxic.

Air pollution is the most crucial from the point of human health. Every human being inhales about 15 - 22 kg of air daily and if that air is polluted, one can easily imagine the damage it can cause to human health.

Source of Air Pollution: The sources of air pollution can be natural or man made.

(i) *Natural sources include*

- Volcanic eruptions emitting poisonous gases like H_2S , SO_2 etc.
- Decay of vegetation,
- Marsh gases

- Pollen grains
- Forest fires
- (ii) *Man made sources include*
 - Increase in population
 - Deforestation
 - Fossil fuel combustion
 - Vehicular emissions
 - Industrialization
 - Use of pesticides, insecticides in agriculture.
 - Explosives used in wars.

Types of air pollutants: The undesirable component in the atmosphere, causing pollution is referred to as pollutant.

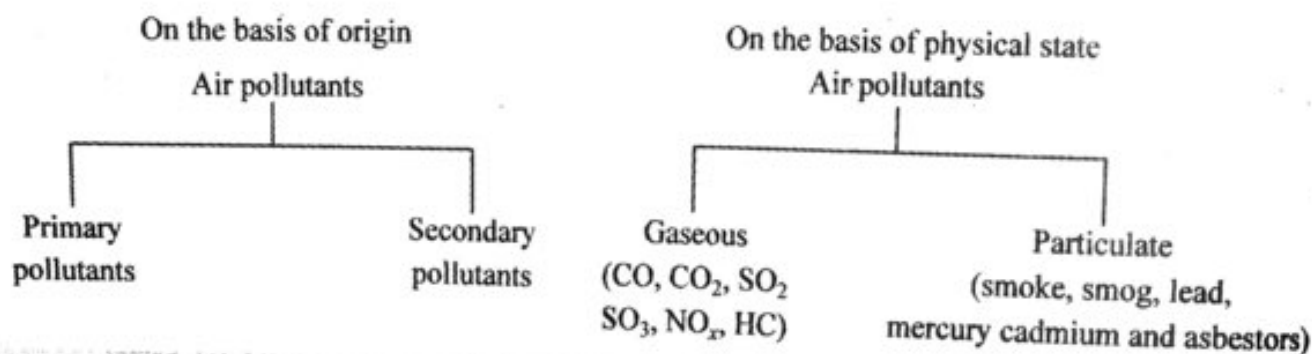
Pollutants can be classified on the basis of origin and physical state. On the basis of origin, pollutants are of two types:

- (i) *Primary pollutant:* Primary pollutants are the products of natural events (like fires, volcanic eruptions) and human activities added directly to air. Examples include, CO, NO₂, SO₂ and hydrocarbons.
- (ii) *Secondary pollutants:* These are formed by the interaction of primary pollutants with each other or with normal components of the air. Examples include ozone, photochemical smog etc.

On the basis of physical state, pollutants are of two types:

- (i) *Gaseous pollutants:* These are the gases which mix with air without settling down. Gaseous pollutants include: CO, CO₂, SO₂, SO₃, NO_x and hydrocarbons.
- (ii) *Particulate pollutants:* Particulate pollutants comprise of finely divided solids or liquids. These include dust, smoke, smog, lead, mercury, cadmium and asbestos.

Classification of Air Pollutants



3.6 SOME COMMON AIR POLLUTANTS

1. Carbon Monoxide (CO)

Carbon monoxide is a colourless, odourless, tasteless gas that is by far the most abundant of the air pollutants.

Source: CO is produced due to:

- incomplete combustion of fuels.
- automobile exhausts. (accounts for 70% of CO in the atmosphere)
- industrial operation

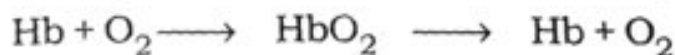
CO is also produced through natural processes such as volcanic activity, natural gas and marsh gas emissions.

Sink for CO: Sink is a system which absorbs/store the pollutant, thereby, nullifying its harmful effect. A large mass of CO is generated and also there is continued increase in CO emissions within recent years. So it is obvious that the amount of CO should also increase in the atmosphere. But it has been found that the amount of CO in the atmosphere remains relatively constant suggesting that a sink or scavenging process also exists in the atmosphere. The micro organisms present in the soil act as a major sink for CO.

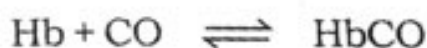
A significant amount of CO is converted into CO₂ by these micro organisms.

Effects: The levels of CO present in the urban air do not effect significantly the plants and materials. However, these levels adversely affect human health. In urban areas, the soil available is insufficient to act as a sink thereby increasing the level of CO beyond permissible limits.

Carbon monoxide interferes with the blood's ability to carry oxygen to different parts of the body. The oxygen combines with haemoglobin, which is also known as oxygen carrier, to form oxyhaemoglobin. This oxyhaemoglobin travels to the different body cells where it gives oxygen to the cell and takes up CO₂ through the lungs.



However, if large amount of CO is present in the surroundings, is inhaled, it readily binds to haemoglobin to form carboxyhaemoglobin. Carbon monoxide, in fact, has a much greater affinity for haemoglobin (about 250 times) than oxygen because it forms a stronger complex with haemoglobin.



Thus the blood (haemoglobin) would carry less oxygen to various parts which inturn affects the brain function. The symptoms include laziness, exhaustion, headache. Acute starvation of oxygen (asyphyxiation) can lead to coma and death.

Physiological effects can be noted even at small percentage of COHb. As the concentration increases, the severity also increases as is evident from the following table 3.2

Table 3.2

Conc. of COHb	Effect
2.5%	Impairment in time-interval discrimination.
5%	Psychomotor response times are affected.
10%	Dizziness and headache
50%	Lethal

The maximum permissible concentrations of CO in the ambient air is 40 ppm for an exposure of 6 - 8 hours. CO concentration in urban areas range from 5 - 50 ppm but on congested red lights the concentration can increase to more than 100 ppm. Cigarette smoke contains more than 400 ppm of CO and smokers frequently have COHb levels between 5 to 10 percent.

Fortunately CO is removed from the blood stream when clean air is inhaled.

Control of CO: Control of the CO pollution can be achieved through the following techniques:

- Modification of engine design:** A low fuel-air ratio reduces NO_x emissions but increase CO emissions. So engine design should be modified so that right proportion of oxygen is maintained for complete oxidation of carbon and hydrogen to CO_2 and H_2O respectively.
- Fuel modification:** The fuels which release lesser amounts of CO are recommended to be used. These fuels include natural gas, methane and blends of light hydrocarbons.
- Treatment of exhaust gases:** Two stage catalytic converters are in use to lower the pollution from exhaust gases. In the first stage NO_x are reduced to N_2 and NH_3 in presence of catalyst such as Pt, Pd and Ruthenium in the presence of reducing gas such as CO. In second stage, oxidizing catalysts of noble metals (Pt, Ru) supported on ceramic materials are used, which ensure oxidation of CO to CO_2 .

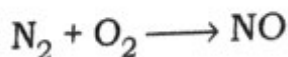
2. Oxides of Nitrogen (NO_x)

A number of oxides of nitrogen such as NO, N_2O , NO_2 , N_2O_3 and N_2O_5 are introduced into the atmosphere due to natural as well as human activity. Out of these, the two oxides NO and NO_2 are responsible for pollution and are considered as pollutants and are represented by NO_x .

NO_2 is reddish brown in colour having pungent smell and is suffocating, whereas NO is colourless and odourless gas.

Source: The source of NO_x include:

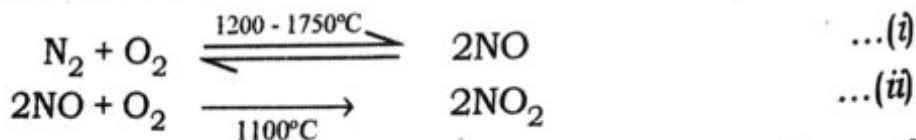
- Natural:** During lightening discharge, N_2 and O_2 in air combine to form NO



NO_x is also formed by the fixation of nitrogen from atmosphere or from nitrogen compounds present in organic matter.

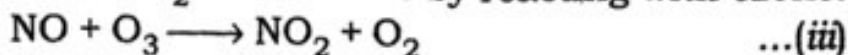
- Man-made sources:**

- NO is formed from N_2 and O_2 at very high temperatures (~ 1210 C to 1765°C). This temperature is usually attained during combustion of fossil fuels in air



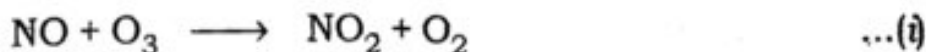
Though the (i) reaction is reversible but rapid cooling of combustion product prevents the dissociation of NO.

The formation of NO_2 by (ii) reaction is quite slow and is not more than 0.5% of the total NO_x present. However, the main reaction through which NO_2 is formed is by reacting with ozone.

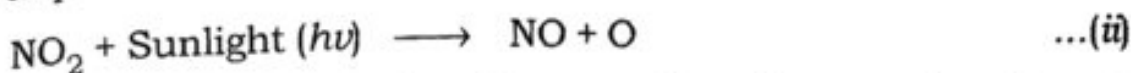


- NO_x is also produced in chemical industries as by products such as in coal based power plants, sulphuric acid and nitric acid manufacturing industries.

Sink for NO_x : The NO and NO_2 undergo various photochemical and chemical reactions in the atmosphere leading to the formation of HNO_3 . Thus NO_2 , NO , pollutants get precipitated as nitrates during rainfall. Ozone plays a significant role in these photochemical reactions.



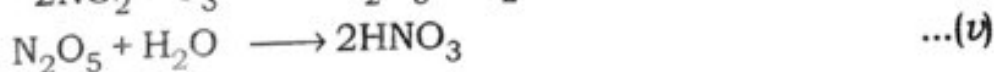
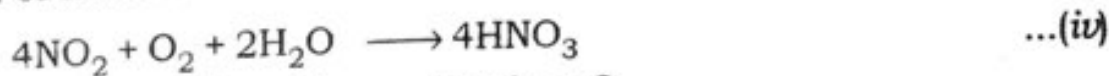
It should be noted that NO remove ozone from atmosphere and this occurs naturally in unpolluted atmosphere.



The nascent oxygen formed in the above reaction then reacts with one of the oxygen molecules producing ozone.



Thus the destruction of NO_2 (reaction ii) and regeneration of NO_2 reaction (i) is a cyclic process. Moreover the amount of ozone should also remain constant as it is used in reaction (i) and regenerated in reaction (iii). However, in the presence of volatile organic compounds, NO_2 follows an alternate path. It reacts with water in presence of oxygen resulting in the formation of nitric acid.



The HNO_3 comes down from the atmosphere to the surface of earth in the form of acid rain. Here it reacts with bases such as ammonia, lime etc. to form nitrates.

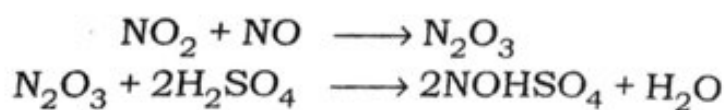
Effects

- Almost all the NO_x emissions are in the form of NO , which has no known adverse health effects at the concentrations found in the atmosphere.
- However NO can oxidize to NO_2 , which in turn may react with hydrocarbons in the presence of sunlight to form *photochemical smog* (discussed later). Photochemical smog is injurious to health.
- The HNO_3 formed by NO_2 causes *acid rain* (discussed later) which has corroding effect on marble and the metallic structures. Moreover, acid rain decreases the pH of the soil, affecting its fertility.

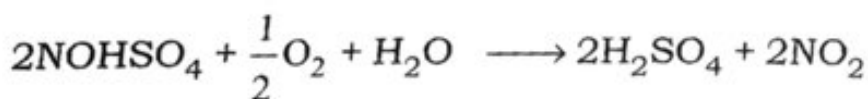
- At higher concentrations than are normally found in the atmosphere, it is an acute irritant. Prolonged exposure to relatively low concentration, has been found to cause bronchitis in children.

Control of NO_x : The control measures for NO_x emissions include the following:

- (a) *Modifying the engine design:* Reducing the amount of excess air for combustion in air helps in controlling NO_x emissions. For instance NO_x decreases from about 600 ppm at 25% excess air to 175 ppm at 1.5% excess air. The burners are so modified that the fuel and air mix more slowly reducing the intensity and temperature of combustion. Two staged combustion process is employed for the purpose.
- (b) *Scrubbing the flue gases:* The flue gases are scrubbed with H_2SO_4 in a scrubber

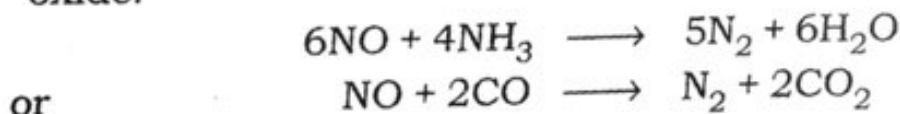


The reaction product obtained from the scrubber is then decomposed.



Excess NO and NO_2 are again circulated through the scrubber.

- (c) *Selective catalytic reduction:* The selective catalytic reduction can be achieved through CH_4 , NH_3 and CO. The NO_x is added to the exhaust gases and mixture is passed over a fixed bed catalyst such as copper oxide.



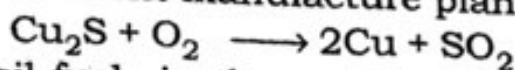
It is interesting to note that the air pollution control methods also require modification of combustion methods. But the combustion process that improve CO emissions tend to make the emissions of NO_x problem worse, and vice versa. For example, to control CO, the air supply should be increased and temperature should be raised so that complete combustion takes place. This however is just the reverse if we want to control the emissions of NO_x which requires just opposite conditions.

3. Oxides of Sulphur

Oxides of sulphur, SO_2 , SO_3 are represented as SO_x . SO_2 is the second most important contributor to air pollution. Though a small concentration of sulphur oxides is desirable for plants and animals but at higher concentrations, it is injurious. Both SO_2 and SO_3 are colourless gases with pungent smell.

Sources

- Natural sources such as volcanic eruptions constitute to about 67% of SO_x pollution.
- Man made sources the sulphur emissions from petroleum refining, copper smelting and cement manufacture plants.

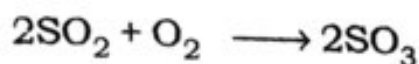


- The burning of fossil fuels in thermal power plants, electric power plants are also one of the main sources of SO_x emissions.



SO_x produced (from the fuel) from natural sources are uniformly distributed all over the globe but SO_x made from man-made sources is more in urban areas.

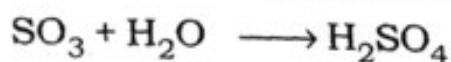
- A small amount of SO_2 undergoes photolytic oxidation in the air to form SO_3



Sinks for SO_x : The sulphur dioxide released in the atmosphere is converted to sulphur SO_3 by photolytic and catalytic processes involving ozone.



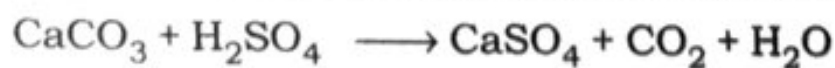
Sulphur oxides combine with water vapour to produce H_2SO_4 aerosol which give rise to the so called "acid-rain" droplets



H_2SO_4 combines with the lime-stone minerals and thus removed from atmosphere. In other words limestone minerals act as a sink for sulphur dioxide.

Effects

- Sulphur dioxide is perhaps the most damaging among the various gaseous air pollutants.
- One of the main causes of 'acid rain' are the oxides of sulphur.
- SO_2 and SO_3 are strong irritants to the respiratory tract. SO_2 is highly water soluble, much more so than any of the other criteria pollutants. As a result, when it is inhaled it is absorbed in the upper respiratory tract.
- If along with sulphur dioxide, the particulate matter in air is also present, the combination becomes detrimental to health. In fact, in every major air pollution episode, this combination has been found to be the major cause leading to death. This combination is suggested as one of the reasons of chronic bronchitis.
- Sulphur dioxide has a very damaging effect on plants. If plants are exposed to SO_3 even at a low concentration (0.03 ppm) for a few weeks, the formation of chlorophyll is slowed down resulting in loss of green colour of leaf. This is called chlorosis.
- Prolonged exposure to sulphur oxide which gets converted to sulphates causes serious damage to buildings, marble, lime stone. The carbonates in these materials are replaced by sulphates.

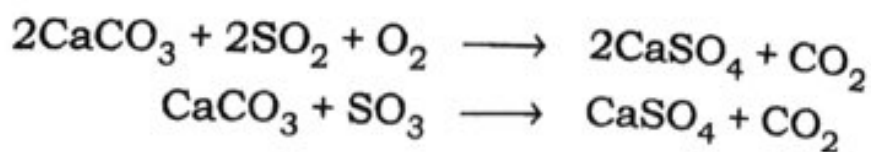


- The CaSO_4 so formed is water soluble and easily washes away, leaving a pitted surface. Many of the world's historic buildings and statues are rapidly being degraded due to this exposure.
- Wool, cotton and leather materials are rotted as a result of the absorption of SO_2 and its oxidation to sulphuric acid, which attacks the proteins in these materials.
 - Paper also absorbs SO_2 which is oxidised to H_2SO_4 causing paper to become brittle and fragile.

Control of SO_x : SO_x can be reduced and controlled by number of ways.

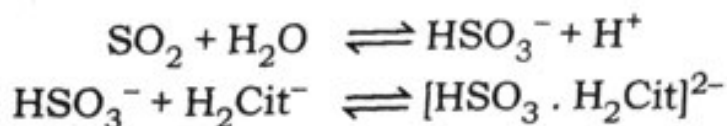
- Use of fuels having low content of sulphur.
- By removing SO_2 from the exhaust gases by using chemical scrubbers. The chemical scrubbers which are used include :

(a) *Lime stone:*



The flue gases (exhaust gases) are passed through the slurry of CaCO_3 which reacts with SO_2 and SO_3 to form CaSO_4 . Though this method is economical but huge amounts of CaSO_4 are produced, the disposal of which is a great problem.

- (b) *Citric acid:* Another method employs the use of citric acid which absorbs the SO_2 from the flue gases. The flue gases are passed through a solution containing citrate ions. The HSO_3^- ions, produced from SO_2 and H_2O , combine with citrate ions to form citrate complex.



The solution containing citrate ion is transferred in a closed vessel and H_2S is passed through it. Sulphur precipitates out and citrate ions are regenerated which are used again.

4. Particulates

Atmospheric particulate matter is defined to be any dispersed matter, solid or liquid, in air. The important characteristics of a particulate are

- Size:** The size of the particulate matter ranges from a diameter of $0.002 \mu\text{m}$ to $500 \mu\text{m}$.
- Concentration:** The density of particulate material vary from few hundred per cm^3 in clean air to 10^5 per cm^3 in polluted air.

A number of terms are used to categorize particulates depending on their size and phase (liquid or solid). Some of these are:

Aerosol: Aerosol is any tiny particle, liquid or solid dispersed in atmosphere.

Dust: Solid particles are called dust if they are caused by grinding or crushing operations.

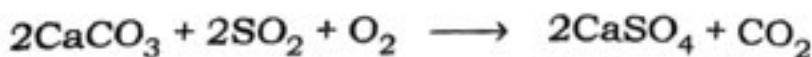
The CaSO_4 so formed is water soluble and easily washes away, leaving a pitted surface. Many of the world's historic buildings and statues are rapidly being degraded due to this exposure.

- Wool, cotton and leather materials are rotted as a result of the absorption of SO_2 and its oxidation to sulphuric acid, which attacks the proteins in these materials.
- Paper also absorbs SO_2 which is oxidised to H_2SO_4 causing paper to become brittle and fragile.

Control of SO_x : SO_x can be reduced and controlled by number of ways.

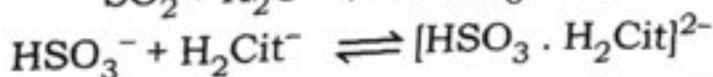
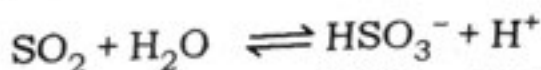
- Use of fuels having low content of sulphur.
- By removing SO_2 from the exhaust gases by using chemical scrubbers. The chemical scrubbers which are used include :

(a) *Lime stone:*



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A number of terms are used to categorize particulates depending on their size and phase (liquid or solid). Some of these are:

Aerosol: Aerosol is any tiny particle, liquid or solid dispersed in atmosphere.

Dust: Solid particles are called dust if they are caused by grinding or crushing operations.

Fumes: Solid particles are called fumes if they are formed when vapours condense.

Mist: Liquid particles are called mist, or more loosely, fog.

Smoke or Soot: These are the terms used to describe particles composed primarily of carbon that results from incomplete combustion.

Fly Ash: These are smaller ash particles coming along with the furnace flue gases and are produced due to combustion of high-ash fossil fuels as in the thermal power plants.

Source: Natural sources include the particulates released from volcanic eruptions, wind and dust storms etc. Man-made sources include the activities such as burning of wood, coal, oil and gaseous fuels, industrial processes, smelting and mining operations. The fly ash emissions from power plants, forest fires etc. release about 450 million tonnes of particulates per year.

Composition: Particulates include organic and inorganic matter. It is estimated that particulate contaminants contain nearly 22 elements. The particulate matter can be organic or inorganic.

Organic Particulates	Inorganic Particulates
Organic particulates arise from automobile exhausts, combustion of fuels. Organic particulates contain polycyclic aromatic hydrocarbons (PAH) which are carcinogenic.	Inorganic particulates include iron oxide, calcium oxide which result from combustion of coal and metallurgical operations. Lead halides are produced in automobile exhausts. Asbestos dust, a fibrous silicate material is introduced by industrial processes. Aerosol mists are generated from oxidations of SO_2 to SO_3 which form H_2SO_4 in presence of water. Cadmium, arsenic, mercury also contribute to the particulate matter.

Effects: The size and chemical characteristics of the particles in atmosphere are more significant than their concentration.

Harmful Effects on Human Life:

The ability of the human respiratory system to defend itself against particulate matter is determined by the size of the particulate matter to a large extent. Large particles that enter the respiratory system can be trapped by lining and hair of the nose. Smaller particles which move to the throat can be captured by mucus membrane and can be removed by spitting. However, smaller particles traverse right upto the lungs and may deposit there. The harmful effects of particulate matter on human life include the following:

- The smaller particles (smaller than $10\text{ }\mu\text{m}$) which are capable of reaching the lungs, accumulate there and act as excellent sites for

adsorption of carcinogenic compounds and may cause lung cancer, bronchial asthma, chronic bronchitis etc. Some of the pollutants, even in small doses, lead to cumulative poisoning i.e. these pollutants, are not excreted once they enter the body. The health effects caused by different types of particulates is summarized in table 3.3.

Table 3.3

Pollutant	Source	Effect on health
Asbestos (fibrous silicate material)	Industry, used as insulation materials in floor and ceilings, fire-proofing, asbestos cement sheets.	Lung disease namely asbestosis, lung cancer known as mesothelioma and other respiratory problems.
Lead	Exhaust from leaded gasoline combustion, lead paint, contaminated soil, plumbing.	Cumulative poisoning impaired mental and physical development, loss of memory, convulsions, coma, inhibits the action of enzymes, causes anaemia, impairs kidney functions.
Silica	Dust containing free SiO_2	Silicosis, exerts a noxious or fibrotic local action in lungs.
Mercury	Spillage from thermometers, thermosets etc. mining and refining of mercury, organic mercurials used in pesticides.	In vapour form it adversely effect the neurological behaviour in humans, mercury poisoning cause brain damage.
Cadmium	Tobacco products, fungicides, cadmium-nickle batteries, fertilizer industries.	Can remain in the body for almost ten years. Lung cancer and kidney damage, bone marrow disorder and cancer. Toxic, carcinogen conjunctivitis, coagulates proteins.
Arsenic	Smoking, pesticides, rodent poisons chemical wastes.	
Fly Ash	Industrial combustion processes	Respiratory diseases, silicosis.

Harmful effects on materials

- Particulates such as fumes, dust, soot, mists and aerosols can bring serve damage to soil, buildings, sculpture and monuments.
- Corrosive particulates accelerates corrosion of the materials.
- Smoke particles stick on every surface of material, forming a film of soot.

Effects on climate

- Particulates play a significant role in maintaining radiation balance and heat balance of earth.

- They provide nuclei for condensation of water vapour leading to the formation of cloud and smoke.
- Particles can absorb free radicals thereby acting as inhibitors for free radical chain reactions.

Control of Particulate Emissions: Many techniques for the control of particulate emissions have been developed. These include:

- Gravity Settling Chamber
- Cyclone Collector
- Filters
- Scrubbers

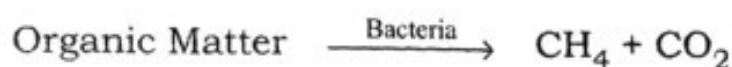
The details are discussed in section 3.7.

5. Hydrocarbons

Many different hydrocarbons are present in air. Most of these are low molecular weight and gases and volatile liquids at ordinary temperatures.

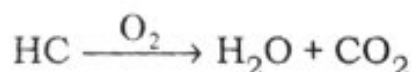
Source

- Natural sources include trees which emit large quantities of hydrocarbons into the air. Methane is the major naturally occurring hydrocarbon emitted in the atmosphere which is produced by anaerobic bacteria through decomposition of organic matter in the soil, water and sediments.



- Man made sources include motor fuels which are mixtures of hydrocarbons. Automobile exhausts emit maximum amount of hydrocarbons in the atmosphere.
- Evaporation of organic solvents used in paints, such as benzene, carbontetrachloride etc. also emit hydrocarbons in air.

Sinks: Hydrocarbons undergo a number of chemical reactions and photochemical reactions through a number of steps and form water and CO_2 in the end.



Effects

- At high concentrations hydrocarbons have carcinogenic effect on lungs.
- Inhalation of benzene, toluene etc. cause irritation in the mucus membrane.
- They are responsible for the of photochemical smog (discussed later)
- Methane causes narcotic effect on human beings.
- Acetylene and propylene at 50 - 500 ppm show toxicity towards plants, damaging their growth.
- Benzpyrine present in tobacco, charcoal etc. induces cancer.

Control

- Hydrocarbons, being thermodynamically unstable tend to get oxidized in the atmosphere by a series of chemical and photochemical reactions. This gives rise to the formation of various end-products such as CO_2 , solid organic particulates and water soluble acids. The solid organic particulates settle down and water soluble acids are washed down by rain.
- The hydrocarbons from vehicular traffic are controlled by the techniques such as incineration, adsorption and absorption.
Adsorption is carried on a bed of carbon or by passing exhaust gases through a liquid in which hydrocarbons will dissolve or become suspended. Incineration or 'after burning' completes the oxidation of hydrocarbon to CO_2 and H_2O .
- Loss of hydrocarbons by evaporation from fuel tanks and carburettor is reduced by installation of a connection system which eventually returns them to the fuel induction system.

3.7 ENVIRONMENTAL EFFECTS OF AIR POLLUTION

The air pollutants discussed above have a great impact on our environment. They are posing a threat to the Earth's general environment. The major environmental effects of air pollution are summarized in fig. 3.3

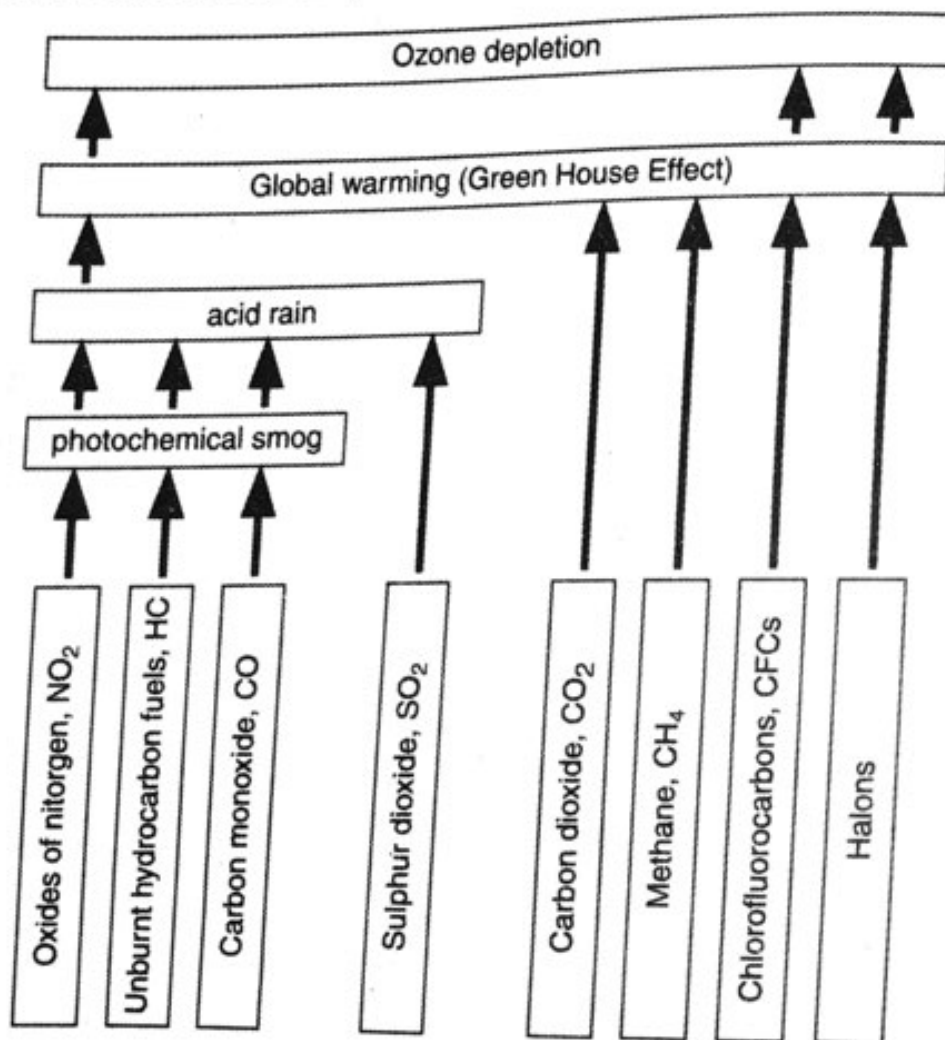


Fig. 3.3 A summary of various environmental problems caused by air pollution.

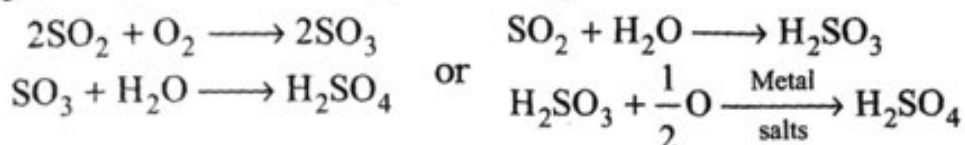
As is evident from the figure that these effects are interrelated and one leads to another. A detailed description of these effects is given below.

1. Photochemical Smog

'Smog' is the combination of two words, 'smoke' and 'fog'. The name was given because fog in the atmosphere condense on the carbon particles of smoke to form smog. However, the smog can have a number of other compositions and methods of formation. Depending upon these factors, smog is of two types:

- (i) **London Smog:** The smog formed from SO_x particulates (soot, ammonium sulphate) and humidity is known as London smog. This type of smog caused the death of around 4000 people in London in 1952, hence the name. Five days of calm foggy weather created a toxic atmosphere which caused bronchitis and acute respiratory problems claiming 4000 human lives. This mixture of smoke, SO_x and fog is chemically a reducing mixture, so it is also termed as *reducing smog*.

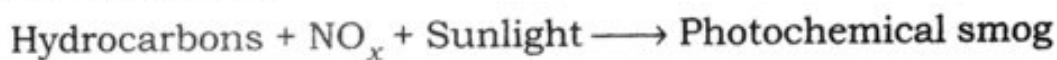
SO_2 in the atmosphere is converted into SO_3 by a number of chemicals present in the particulates. SO_3 combines with water in the atmosphere forming a fog of sulphuric acid droplets. These droplets condense on the carbon particles of soot and are drawn into the respiratory tract during breathing.



London smog is formed in the early morning hours of winter months. After sunrise it increases due to the oxidation of SO_2 to SO_3 and subsequent combination with water to form sulphuric acid aerosol.

- (ii) **Los Angeles Fog (Photochemical Smog):** Photochemical smog results from the photochemical reactions of the atmosphere. Photochemical smog is the atmospheric haze that is formed near many large cities and it is due to the action of sunlight on the hydrocarbons and the nitrogen oxides. In this type of smog, the main unhealthy ingredient is 'ground level ozone'. This type of smog was first observed in Los Angeles in 1950 and hence is named as 'Los Angeles smog'.

Composition: In 1951, Haagen-Smit observed a key mechanism involving an atmospheric reaction between unburned gasoline-range hydrocarbons (from unburned automobile emission) and nitrogen dioxide in presence of sunlight, a photochemical reaction. The products were peroxyacyl nitrates (PANs) and ozone. In very simplest terms, we can express the formation of photochemical smog as



Photochemical smog is mixture of substances formed by the reaction of hydrocarbons, NO_x and sunlight.

Photochemical smog is an *oxidizing smog* and should be clearly distinguished from the usual reducing 'smog' which is formed due to the combination of smoke and fog.

Thus, the formation of photochemical smog requires nitrogen oxides ($\text{NO} + \text{NO}_2$), volatile organic compounds (hydrocarbons) and ozone which lead to the formation of peroxyacetyl nitrate (PAN). An account of the source and effect of these chemical pollutants is given in table 3.4

It is worthwhile to mention that PAN was not detected until recognized in smog. It has higher toxicity to plants than ozone.

Table 3.4: Major chemical pollutants in photochemical smog: Source and environmental effects.

Toxic Chemical	Source	Environmental effects
Nitrogen oxides (NO and NO₂)	• combustion of oil, coal, gas in both automobiles and industry • bacterial action in soil • forest fires • volcanic action • lightning	• decreased visibility due to yellowish colour of NO₂ • NO₂ contributes to heart and lung problems • NO₂ can suppress plant growth • decreased resistance to infection • may encourage the spread of cancer
Volatile organic compounds (VOCs)	• evaporation of solvents • evaporation of fuels • incomplete combustion of fossil fuels • naturally occurring compounds like terpenes from trees	• eye irritation • respiratory irritation • some are carcinogenic • decreased visibility due to blue-brown haze
Ozone (O₃)	• formed from photolysis of NO₂ • sometimes results from stratospheric ozone intrusions	• bronchial constriction • coughing wheezing • respiratory irritation • eye irritation • decreased crop yields • retards plant growth • damages plastics • break down rubber • harsh odour
Peroxyacetyl Nitrates (PAN)	• formed by the reaction of NO₂ with VOCs (can be formed naturally in some environments)	• eye irritation • high toxicity to plants respiratory • damaging to proteins irritation

Development of Photochemical Smog (Conditions required):

For photochemical smog to occur there are four necessary conditions:

- presence of nitrogen oxides
- sunlight
- hydrocarbons
- temperature above 18°C

In cities, early morning traffic increases emission of NO_x and hydrocarbons. These NO_x and hydrocarbons (volatile organic compounds) react and form NO₂. NO₂ breaks down forming by products and atomic oxygen. Atomic oxygen combines with the molecular oxygen to form ozone.

NO_2 also reacts with VOC to produce toxic chemicals such as PAN. In the evening hours, when the sun goes down, the production of ozone is halted. The ozone in the atmosphere is consumed by different reactions.

Chemistry of Photochemical Smog

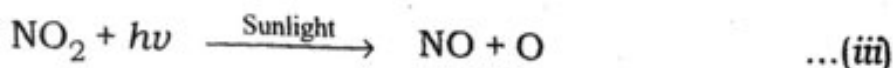
Photochemical smog formation proceeds through a sequence of reactions, all involving a free radical mechanism. The main steps can be outlined as follows:

- Nitrogen dioxide is formed as a result of one of the following reactions. At high temperature of the combustion chamber of the vehicle N_2 and O_2 combine to form NO



The (i) reaction is endothermic so the formation of NO is favoured at high temperatures.

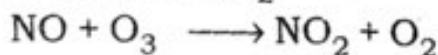
- NO_2 so formed is an efficient absorber of ultra-violet (UV) radiations. Thus UV rays in the sunlight cause the photodissociation of NO_2 to form NO and atomic oxygen



- The atomic oxygen formed in (iii) then reacts with abundant oxygen molecules producing ozone



Ozone can convert NO back to NO_2



Note that nitric oxide (NO) acts to remove ozone (O_3) from the atmosphere.

This set of reaction creates a cycle as represented in Fig. 3.4

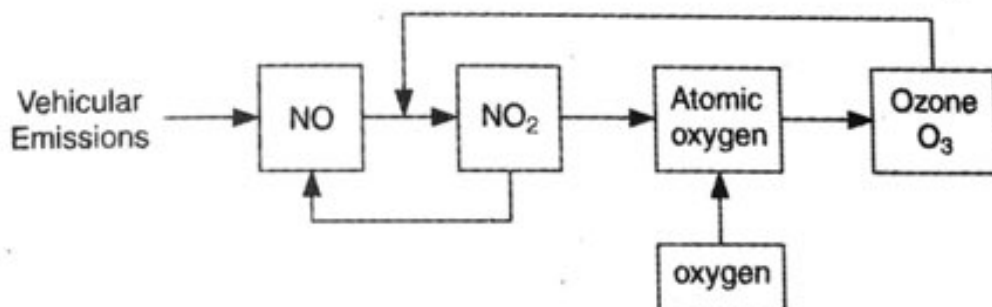
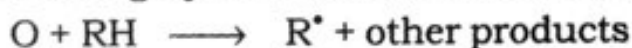


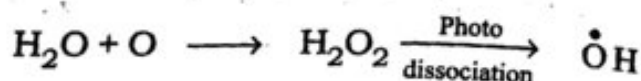
Fig. 3.4

Thus NO and O_3 produced are used up and no extra NO_2 is added up to the atmosphere. But when hydrocarbons are present, then the chain propagation reactions take place.

- The hydrocarbons combine with the oxygen atom produced by the photolysis of NO_2 , to form highly reactive intermediates (free radicals)



Oxygen atom can also react with H_2O to form hydroxyl radicals.



- The free radicals then react with O_2 or NO to produce a number of by products which are highly reactive organic oxidants i.e. secondary pollutants such as acrolein, formaldehyde, peroxyacetyl nitrate (PAN). These secondary pollutants collectively form photochemical smog.

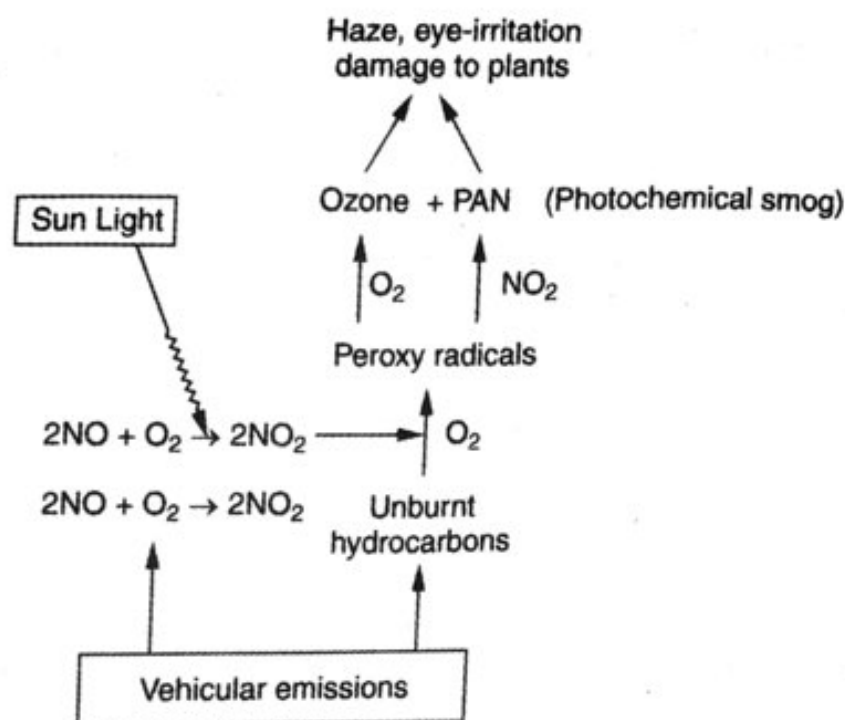
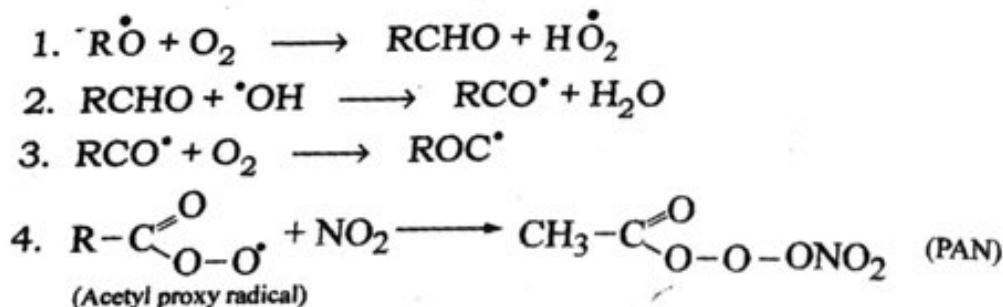
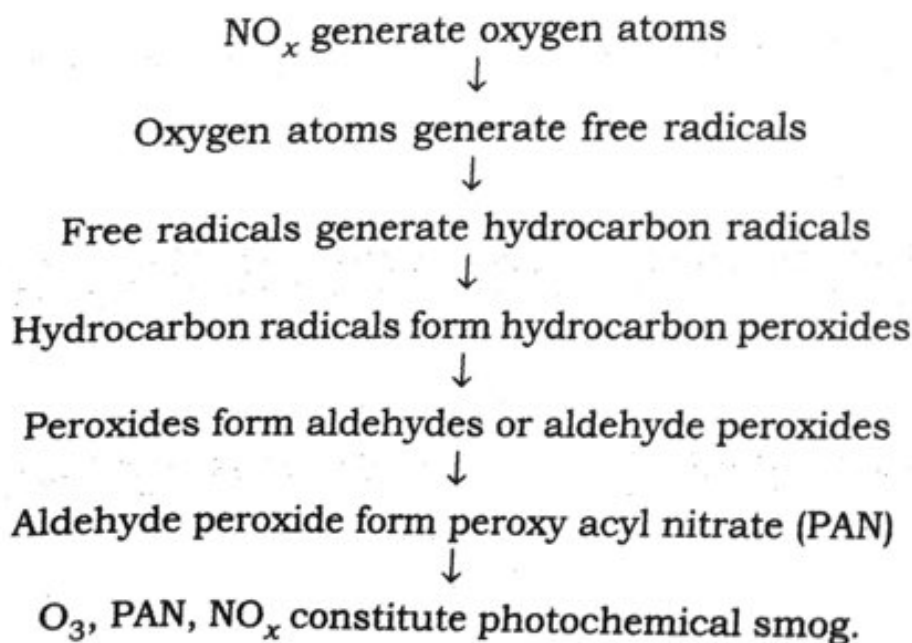


Fig. 3.5 Formation of photochemical smog.

To summarize, the sequence of photochemical smog can be represented as -



All these compounds (PAN, aldehydes, ketones, ozone) produce a number of respiratory problems and irritation in the eyes as is discussed in table 3.4.

The word 'smog' is misnomer here because smog is a combination of smoke and fog and neither of these are present in photochemical smog which is a mixture of NO_2 , O_3 , PAN, aldehydes, and hydrocarbons. The difference between London smog and photochemical smog are tabulated in table 3.5.

Table 3.5

Classical smog (London smog)	Photochemical smog (Los Angeles smog)
1. It involves smoke and fog. 2. This type of smog was first observed in London in 1952. 3. It is formed due to pressure of SO_2 and humidity in the air which combine to form H_2SO_4 fog which deposits on the particulates. 4. It is formed in the months of winter particularly in the morning hours when the temperature is low. 5. It is a reducing in character.	1. This type of smog was observed in Los Angeles in 1950. 2. It is formed due to photochemical reaction taking place when air contains NO_2 and hydrocarbons. 3. It does not involve any smoke or fog. The word smog is a misnomer here. 4. It is formed in the months of summer during afternoon when there is bright sunlight so that photochemical reactions can take place. 5. It is oxidizing in character.

2. Acid Rain

Unpolluted rain water is slightly acidic due to the presence of carbondioxide in air. The CO_2 combines with water droplets to form a weak solution of carbonic acid with a pH of about 5.6



However, in polluted environments, the rain passes through an atmosphere polluted with oxides of sulphur (SO_x) and nitrogen (NO_x). The falling rain water reacts with these oxides to form a mixture of sulphuric acid and nitric acid and water. This is known as acid rain. Therefore rain water with pH value lower than 5.7 is called acid rain. In some parts of the world, a pH as low as 2.5 in rainwater has been recorded.

The term acid precipitation is often used to specifically describe wet forms of acid pollution that is found in rain, snow, fog, cloud and vapour. The term 'acid rain' was used by Angus Smith.

It is more correct to refer to acid rain as *acid deposition* because it may be exhibited in more physical states than rain alone.

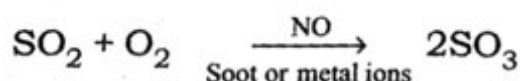
Actually acid rain is the one phase of acid deposition which can either be wet or dry.

Wet deposition: Occurs in the atmosphere and falls on earth as rain, snow, dew etc.

Dry deposition: Dust particles containing sulphates and nitrates, settled on earth, is called dry deposition.

Acid Rain Formation: It is well known that oxides of nitrogen and sulphur are present in the atmosphere. The sources of these oxides have been already discussed earlier in the chapter:

- The oxides of sulphur react with moisture to form H_2SO_4 in sequence of steps -

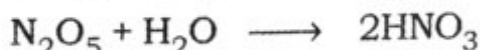
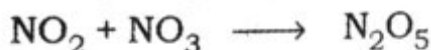
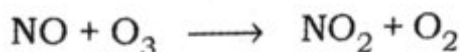


This reaction is catalysed by aerosol containing metal ions of Cu(II) , Fe(II) and oxides of Pb etc. Even soot particles are also known to catalyze the reaction.

The SO_3 formed reacts with water vapour in air to form H_2SO_4 .



- Similarly the oxides of nitrogen also undergo a series of photochemical and chemical reactions to form HNO_3 .



These acid droplets are partly neutralised with bases such as particulate lime, NH_3 etc.

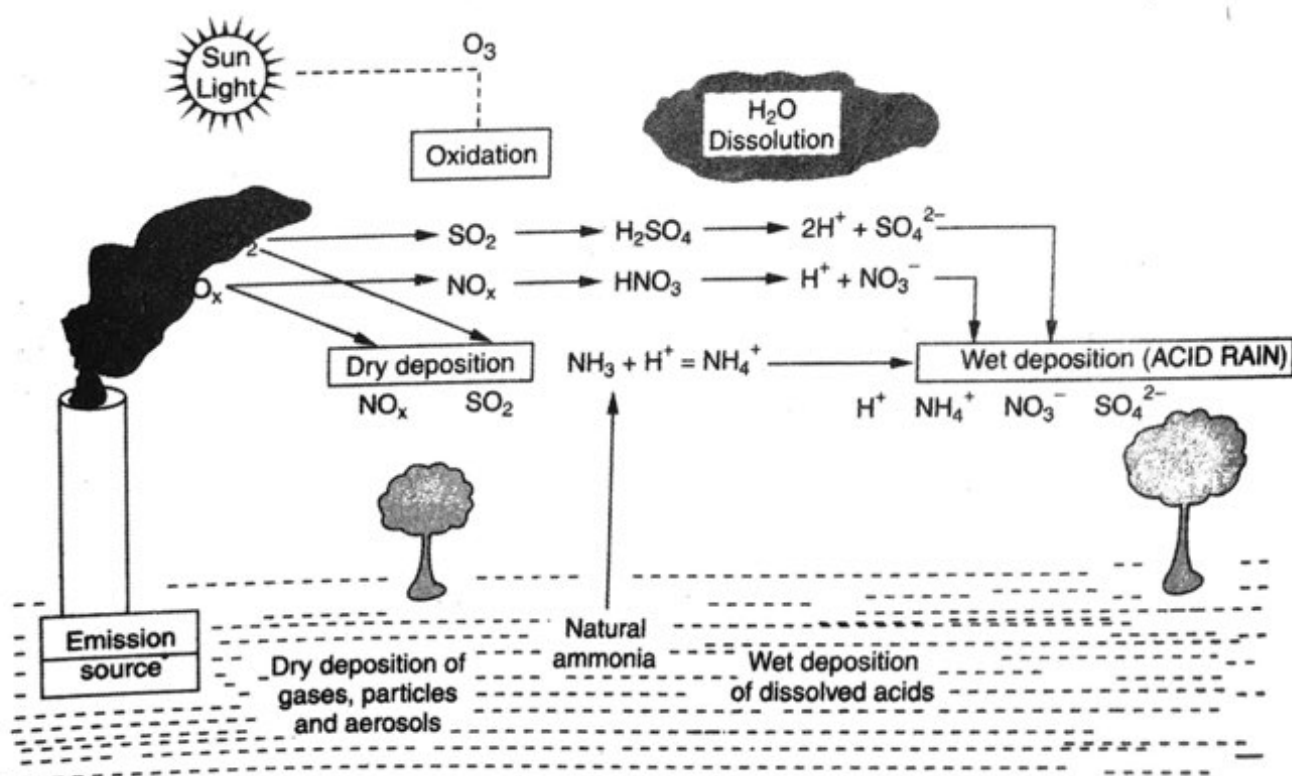


Fig. 3.6 Formation of Acid Rain.

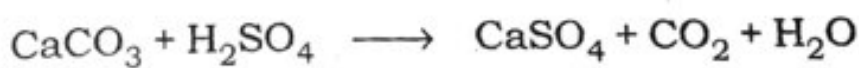


- These salts and remaining H_2SO_4 , HNO_3 droplets alongwith HCl , which is released into the atmosphere by man made and natural HCl emissions, give rise to acid precipitation, known as 'Acid Rain'. H_2SO_4 contributes maximum followed by HNO_3 which in turn is followed by HCl . Fig. 3.6 illustrates the formation of acid rain.

Effects of Acid Rain: Acid rain represents one of the major consequences of air pollution. Longer the SO_x and NO_x remain in the atmosphere, greater are the chances of their oxidation to H_2SO_4 and HNO_3 by various photochemical and catalytic chemical reactions. Acid rain can occur at a place far away from the source of pollutants.

Acid rain influence the environment in several different ways. Some of the harmful effects are discussed below:

- **Acidification of lakes and streams:** Acid rain increases the acidity of water in lakes and streams which is detrimental to fish life. At pH 5, most fish cannot hatch and at lower pH some fish die. In some lakes, acid rain has completely eliminated fish species. Moreover as acid rain flows through soils, aluminium is released from soils into the lakes. Both low pH and increased aluminium levels are toxic to fish.
- **Damage to the vegetation:** Acid rain does not usually kill trees directly. It causes damage due to:
 - damaging their leaves
 - leaching the nutrients available to the plant
 - exposing the plants to toxic substances slowly released from soil.
 - alteration of seed germination characteristics.
- **Damage to building and sculptural material:** The buildings and materials made of marble, lime-stone, mortar etc. suffer extensive damage due to acid rains. The reaction with marble takes place as



The fumes from Mathura refinery are responsible for causing acid rain in Agra, attacking the Taj Mahal. Paints on car can react with acid rain causing fading.

- **Effect on human:** Acid deposition can effect human being life by releasing toxic metals such as mercury and aluminium through the acidification of soil. These toxic metals then end up in drinking water, crops and fish. Aluminium is believed to cause Alzheimer's disease. Moreover, high amount of acidic pollution show increased frequencies of chest colds, allergies and coughs.

3. Greenhouse Effect

The greenhouse effect is the rise in temperature that the earth experiences because certain gases in the atmosphere (H_2O vapour, CO_2 , NO and methane) absorb energy from the sun. Without these gases, heat would escape back into space and earth's average temperature would be about 60°C colder.

The phenomenon has been named as green house effect, because it is similar to heat trapping effect of the glass walls in a horticultural green house. The sun's energy enter through the glass. This energy is absorbed by the soil and other structures of green house and is remitted in longer wavelengths. The longer wavelengths cannot pass through the glass and are retained in the green house making it warm and cosy for the growth of the plants. Similarly in the atmosphere the CO_2 , H_2O , methane etc. play the role played by glass in green house and hence these gases are termed as green house gases.

The Process: The earth receives a tremendous quantity of radiant energy from the sun, about 30% of which is reflected into space by the earth's atmosphere. The remaining energy passes through the atmosphere to the earth's surface. Some of this energy is absorbed by plants to drive photosynthesis and some by the oceans to evaporate water, but most of it is absorbed by soil, rock, and water to increase the temperature of the earth's surface. *This energy is in turn radiated from the heated surface mainly as infrared radiation, often called heat radiation.*

However, only a small portion of this energy actually makes it back to space. The majority of the outgoing infrared radiations, which have longer wavelengths than the solar radiations, are absorbed by the greenhouse gases present in the atmosphere. Thus CO_2 , H_2O strongly absorb infrared radiation and radiate it back into the atmosphere towards the earth. (Fig. 3.7). A net amount of thermal energy is retained by the earth and makes it warmer than it would be without these gases in the atmosphere. The average temperature of the earth's surface is 288 K. It would be -255 K without the greenhouse gases. Without the greenhouse effect, life on this planet would probably not exist as the average temperature would be a chilly -18°C.

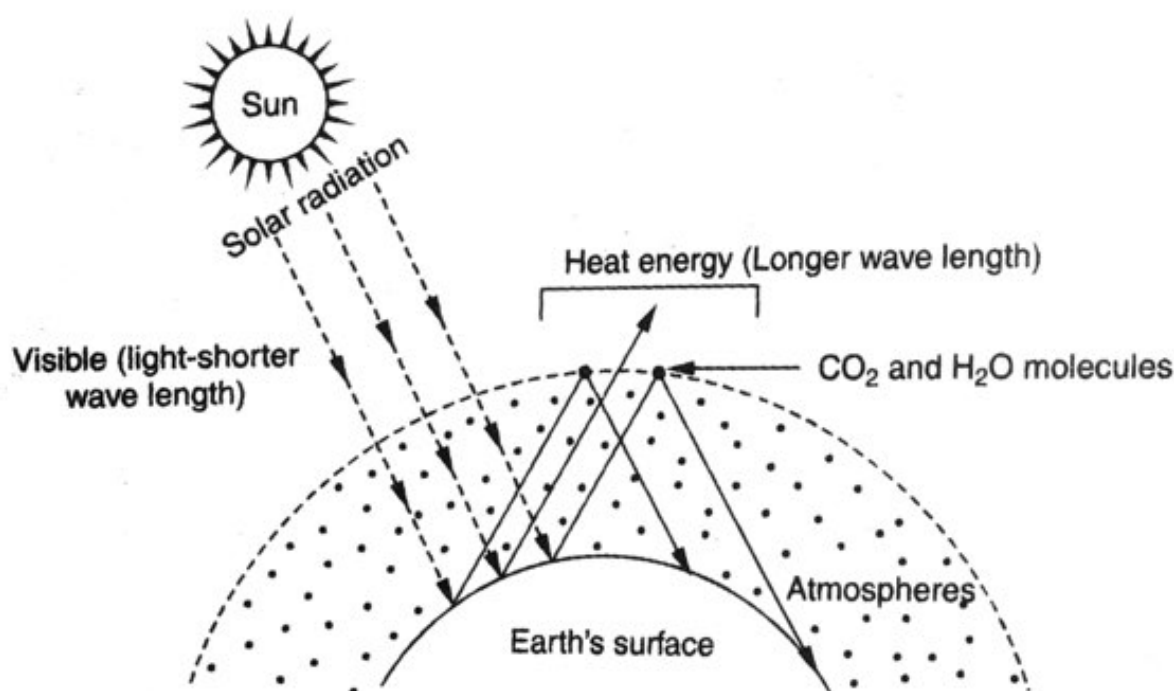


Fig. 3.7 The Visible Light Strikes the Earth and part of it is transforms into Infrared Radiation. Infrared Radiation is strongly absorbed by CO_2 , H_2O , CH_4 and N_2O in the Atmosphere keeping the Earth warmer.

Scientists use the term *albedo* to define the percentage of solar energy reflected back by a surface.

Green House Gases: The four major green house gases are carbondioxide (CO_2), methane (CH_4), nitrous oxide (N_2O) and chloroflourocarbons (CFCs). The CO_2 is the major contributor to green house effect as in shown in fig. 3.8

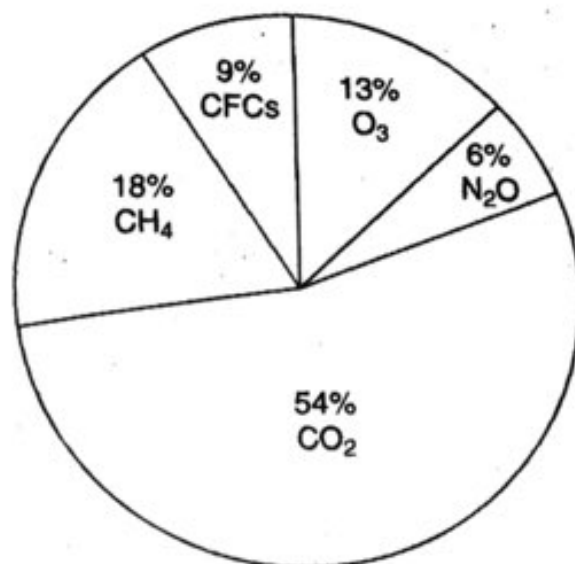


Fig. 3.8 Contribution of atmospheric gases to greenhouse effect.

Enhanced Green House Effect (Global Warming)

The green house has been described as a natural phenomenon that is responsible for life on earth. Natural greenhouse effect is good for human life. However, due to industrial revolution, the concentration of green house gases is increasing constantly as is apparent from table 3.6.

Table 3.6: Past and present concentration and sources

Greenhouse Gas	Concentration 1750 (at the beginning of industrial revolution)	Present concentration	Percent increase	Natural and anthropogenic source
Carbon Dioxide	280 ppm	360 ppm	29%	Organic decay; Forest fires; Volcanoes; Burning fossil fuels; Deforestation; Land-use change
Methane	0.70 ppm	1.70 ppm	143%	Wetlands; Organic decay; Termites; Natural gas & oil extraction; Biomass burning; Rice cultivation; Cattle; Refuse landfills

Greenhouse Gas	Concentration 1750 (at the beginning of industrial revolution)	Present concentration	Percent increase	Natural and anthropogenic source
Nitrous Oxide	280 ppb	310 ppb	11%	Forests; Grasslands; Oceans; Soils; Soil cultivation; Fertilizers; Biomass burning; Burning of fossil fuels
Chlorofluorocarbons (CFCs)	0	900 ppt	Not applicable	Refrigerators; Aerosol spray propellants; Cleaning solvents
Ozone	Unknown	Varies with latitude and altitude in the atmosphere	Global levels have generally decreased in the stratosphere	Created naturally by the action of sunlight on molecular oxygen

As a result of higher concentration of the green house gases, scientists predict that the earth's atmosphere will be warmer and will lead to global warming. The global warming will have drastic effect on climate and may even prove even disastrous to human life. That is why, we often hear about green house effect in somewhat negative terms. But the negative concerns are related to possible impacts of an *enhanced* green house effect. Some scientists fear that if proper precautions are not taken, the concentration of the green house gases in the atmosphere may double in next 50 years leading to increase in global temperature by 4 to 5°C.

Effects of Global Warming: Global warming is one of the greatest challenges of 21st century:

- Predicting the amount of warming is accomplished by computer modelling. Computer model suggest that doubling of CO₂ may raise the temperature between 1 and 3°C. Some scientists estimate that the earth's average temperature has already increased by 0.3 to 0.6°C.
- The increased temperature would lead to melting of glaciers and polar ice caps and the level of sea level may rise thereby flooding the coastal lands and change the climate radically.
- Even 1.5°C raise in surface temperature can adversely affect the food production in the world.
- Changing regional climates could alter forests, crop yields and water supplies.

Remedial Measures for Global Warming

- Reducing consumption of fossil fuels such as coal and petroleum.
- Recovering green house gases from the atmosphere.
- Reduction in CFC production
- Reforestation and conservation of forests, so that plants can take up CO_2 .
- Development of environmentally compatible technologies.

4. Ozone Layer and Its Depletion

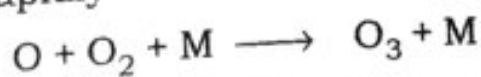
Ozone is a very small part of our atmosphere but its presence is nevertheless vital to human well-being.

Most ozone resides in the upper part of the atmosphere, called stratosphere extending from 16 km to 40 km. About 90% of ozone is contained in this 'ozone-layer' and remaining 10% ozone is present in troposphere. The high concentration of ozone (about 10 ppm) in stratosphere shields us from harmful ultraviolet light from sun and hence is referred to as earth's protective umbrella.

Ozone is naturally present in our atmosphere. It is formed in the atmosphere in multistep chemical processes that require sunlight. Oxygen which make up 21% of the atmosphere undergo photolytic decomposition in presence of sunlight.

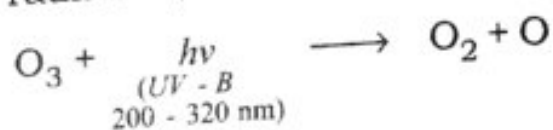


The atomic oxygen rapidly reacts with molecular oxygen to form ozone



where M represents a third body (N_2 or O_2) necessary to carry away the energy released in the reaction.

Stratospheric ozone is considered good ozone because it absorbs ultraviolet (UV - B) radiation, which are harmful for human beings.



The above combination of reactions form a long chain in which oxygen atoms are constantly being shuttled back and forth between the various molecular forms.

Other UV radiation, UV-A is not absorbed significantly by ozone, which reaches the surface of earth. However UV-A is not carcinogenic at usual exposure level of earth. Fig. 3.9 shows the two bands designated as UV-A and UV-B.

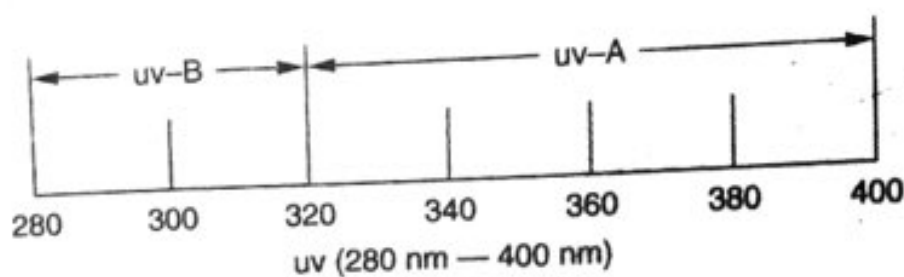


Fig. 3.9 UV bands.

Effect of Ozone Layer

- As is discussed ozone layer protects us from harmful UV-B rays absorbing these radiation. In absence of ozone layer these rays would reach earth and cause (i) skin-cancer (ii) damage to the plants (iii) sun burns (iv) melanoma (v) leukaemia (vi) cataracts etc.
- The absorption of UV-B radiation by ozone is a source of heat in stratosphere. As a result ozone plays a key role in maintaining the temperature structure of earth's atmosphere.

Bad ozone (Tropospheric ozone): Ozone is also formed near earth's surface by chemical reactions caused by presence of pollutants. This ozone is considered bad because more ozone comes in contact with humans, plants and animals. Ozone strongly reacts to destroy or alter many other molecules. In humans excessive exposure to ozone can lead to chest pains, throat irritation and cough. It also contributes to global warming. It also aids in the formation of photochemical smog.

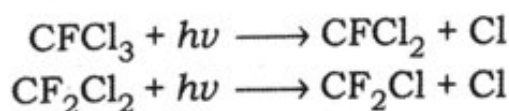
Depletion of Stratospheric Ozone: It was discovered in the mid-1970s that some human activities could destroy ozone and deplete the ozone layer. The emission of ozone depletion gases is a cause of great concern. Ozone depletion has been found to occur through following types of compounds:

- Chloroflouro carbons (freons)
- Nitric oxide
- Reactive hydroxy radicals
- Atomic oxygen

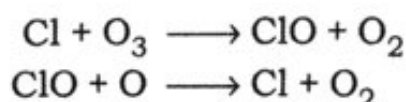
Chloroflouro carbons (CFCs): These are the compounds containing chlorine and fluorine. These are introduced into the atmosphere by aerosol sprays in which they act as propellants, refrigerating equipments in which they act as coolants, and in plastic foams such as 'thermocole' or styrofoam.

CFCs are insoluble in water and are not washed out through rain. Slowly they diffuse into stratosphere, which may take several decades.

The CFCs such as CFCl_3 (Freon-II) and CF_2Cl_2 (Freon-12), absorb UV radiation resulting in photodissociation to give chlorine atoms.



The free Cl now reacts with ozone in a catalytic manner,



Thus Cl is regenerated and can attack other ozone molecules. ClO can react with ozone *six times faster* than any of the nitrogen species, it becomes the dominant mechanism to destroy stratospheric ozone. It has been found that one molecule of CFC can destroy one lakh of O_3 molecules in the stratosphere. Due to depletion of ozone by CFCs it was found in eighties

that a large hole has been created in the ozone layer. That is why the use of CFCs has been completely banned.

Nitric Oxide: The main source of nitric oxide in the atmosphere is nitrous oxide N_2O produced through bacterial action of the surface of the earth. It gradually crosses to the stratosphere and reacts with ozone



Net reaction



Thus in each cycle two molecules of ozone are destroyed to yield three molecules of oxygen.

Reactive hydroxy radicals: The hydroxy radicals enhance the reactions of CFCs and NO with O_3 . $\dot{O}H$ reacts with ozone as follows:



Effect of Ozone Depletion

- The existing ozone layer screens out more than 99% of incoming *UV* radiations. The small fraction that gets through is known to cause sunburn, skin cancer, and various kinds of damage to animals and plants.
- The genetic material DNA has the capacity to absorb *UV* radiations which may cause various disruptive effects.
- A 5% depletion in ozone produces 10% increase in radiations reaching the earth according to an estimate. That is, it would produce 5 times additional skin cancer per year worldwide.
- Increased *UV* radiations impairs the growth of certain crop plants and adversely affects a wide variety of organisms.
- Exposure to *UV* radiations damages the cornea and lens of eye causing blindness.
- Ozone depletion has a strong effect on climate. Ozone depletion itself is not the principal cause of climate change. However because ozone is a greenhouse gas, ozone changes and climate change are linked in important ways.

Substitute for CFCs: CFCs are found to be the major culprit in ozone depletion hence the use of CFCs has been totally banned worldwide.

Montreal Protocol, an international agreement on substances that deplete the ozone layer, established legally binding controls for about 185 nations on the production and consumption of ozone depleting gases.

Thus substitutes for CFCs were worked out and these days hydrochloroflouro carbons (HCFC) are being substituted for CFCs in air

conditioning and refrigerating units. Because HCFC are chemically removed primarily in the troposphere and hence do not cause ozone depletion inspite of chlorine atom present. Hydroflouro carbons (HFCs) are also used as substitute compounds for CFCs. HFCs contain no chlorine atom they do not contribute to ozone depletion. However, these may contribute to global warming.

The ozone layer is expected to recover by the middle of 21st century if nations comply with Montreal Protocol.

3.8 AIR POLLUTION CONTROL

The most effective method of dealing with air pollution problem is to prevent the formation of pollutant or to reduce their emissions at the source itself. Some of the industrial processes may be suitably modified so as to minimize the extent of pollution.

Main routes to cutting pollutant emissions are:

- (i) Reducing the consumption of fossil fuels.
- (ii) Better designing of the equipment to control emissions.
- (iii) Removing pollutants from exhaust and gases (source correction methods)

Based on the above points various pollution control methods have been adopted. Some of which are discussed below;

1. Source Correction Methods: Air pollution can be minimised at source by taking the following measures:

- (a) *Substitution of the raw material:* A simple control of air pollution is to use pure-grade raw material for combustion. For example, use of low-sulphur oil and coal is recommended.
- (b) *Modification in the process:* It involves changes and modification in industrial technologies aimed at reducing the atmospheric emissions of pollutants. For example, the use of exhaust hoods and ducts over several types of industrial ovens allow the recovery of various solvents that could have become air pollutants. H_2S emissions can be recycled and used to recover sulphur.
- (c) *Alternations in equipment:* By suitably modifying the carburation and ignition systems in an automobile engine, air pollution can be reduced. For example, the basic oxygen furnace which are replacing open hearth furnace in steel industry pose much less air pollution problems. Evaporation from petroleum refineries can be minimised by suitable design modification of the tanks.

2. Air pollution control devices: Even if the above measures are taken, in certain cases emissions of pollutants cannot be prevented and pollution control devices are employed depending upon the type of pollutant (gaseous or particulate).

(i) **Devices for controlling Particulate Emissions:** Particulate matter in gaseous effluents consists of discrete and minute particles of sizes ranging from $0.1\ \mu\text{m}$ to $100\ \mu\text{m}$. The device selection is based upon physical and chemical characteristics of particulates, size, temperature and humidity. The equipments presently available use one of the following techniques:

(a) **Gravitational Settling:** This technique is generally employed to remove large particulates ($> 50\ \mu\text{m}$) from the gaseous effluents. In this the velocity of the horizontal carrier gas is reduced adequately so that the particles settle by gravitational force. The collector efficiency is improved by dividing the setting chambers by a number of trays. (Fig. 3.10)

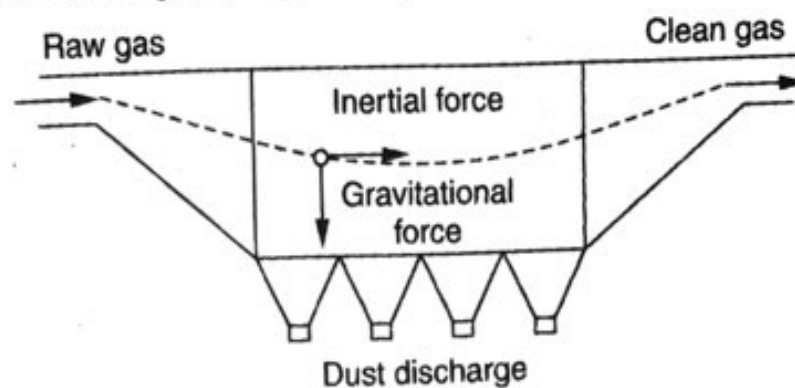


Fig. 3.10 Gravitational settling chamber

(b) **Cyclonic separators:** In this method, gas containing particulates is allowed to flow into a tight circular spiral-fitted chamber. Now the particulates present in the gaseous stream possess greater inertia than the gas molecules and experience greater centrifugal force, which drive these particulates towards the wall of the chamber. From there they settle down due to force of gravity. These type of cyclone separators are used in power plants and industries dealing with rock products. (Fig. 3.11)

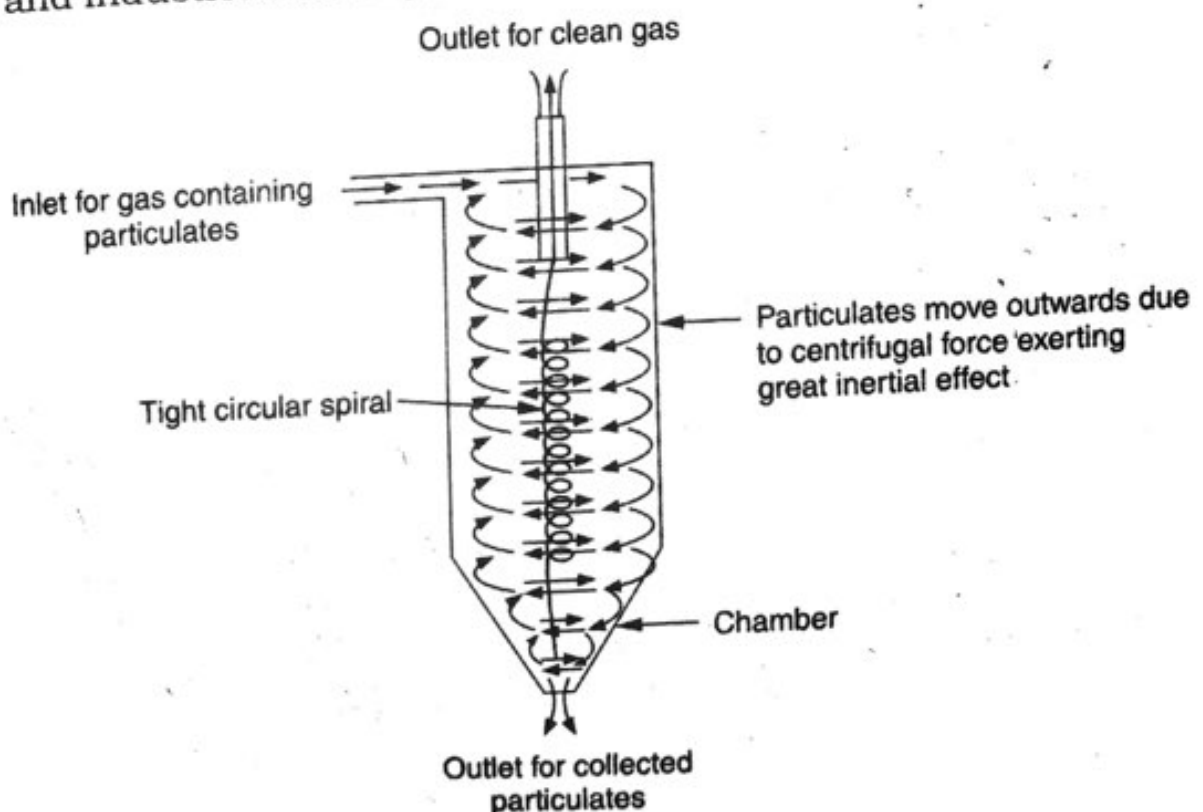


Fig. 3.11 A cyclone collector.

The cyclone collector is most efficient for the removal of large particles of size ranging from 5 - 20 μm (smoke, dust etc.). This technique is more efficient than gravitational setting because centrifugal force on particles in a spinning gas stream is much stronger than the force of gravity.

- (c) *Fabric filters:* Small solid particles are removed effectively by the use of fabric filters. The exhaust gases are forced through bags of very fine cloth-like mesh which are capable of retaining small particles. Fabric filters generally consist of cotton, wool, nylon, dacron or coated glass cloth. Fabric filters are used in industries dealing with rock products, pigments etc. (Fig. 3.12).

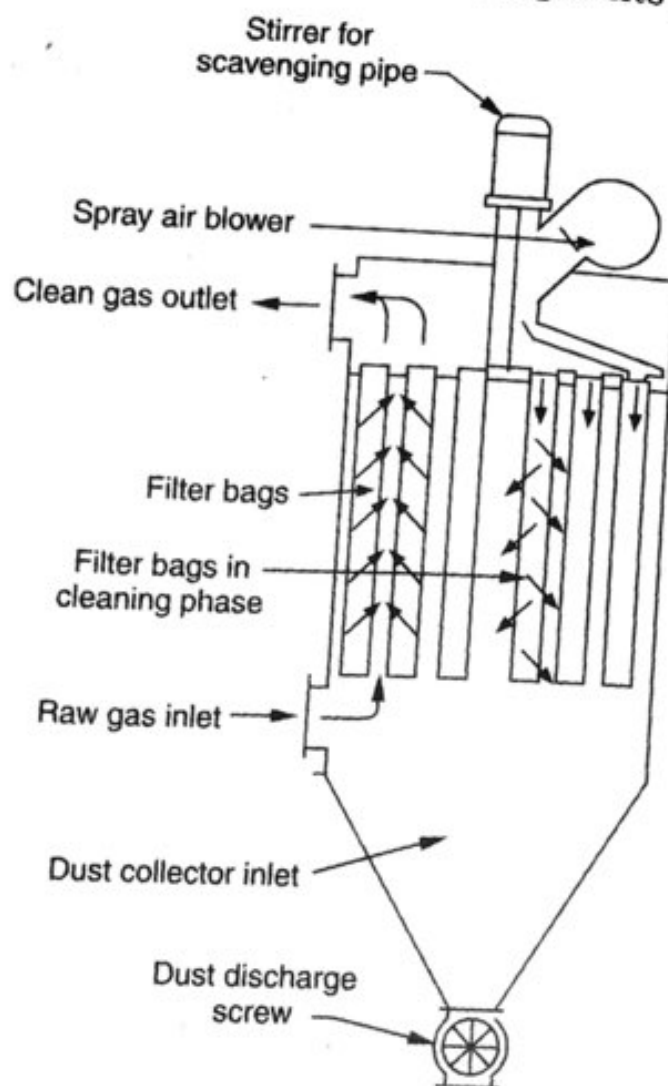


Fig. 3.12 Fabric filters.

- (d) *Electrostatic precipitator:* The most efficient and widely used device for controlling particulate emissions in power plants, cement and paper industry and oil refineries is the electrostatic precipitator. In this technique the particles present in gaseous exhaust are charged electrically and separated from the gas stream under the influence of electric field.

The electrostatic precipitator consists of a series of grounded thin plates which are charged to high voltages, alternately + and -. Particles approaching these plates (around 50,000 V) become charged and are attracted towards opposite plates. From these plates they fall into a collector (hopper) below (Fig. 3.13). The efficiency of electrostatic precipitator is 97.9% and it is

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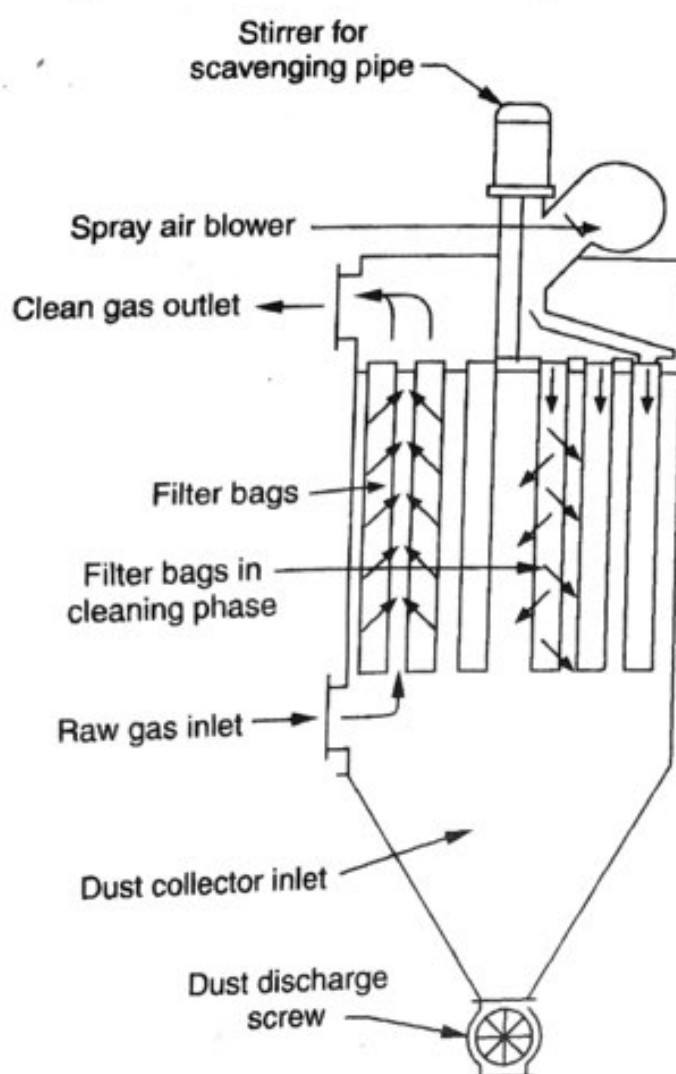


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capable of cleaning 150,000 litres of gas per minute at a temperature of 600°C.

Electrostatic precipitators are the devices of choice and are widely used in power plants, paper and pulp industries, chemical industries such as sulphuric acid plants, iron and steel plants.

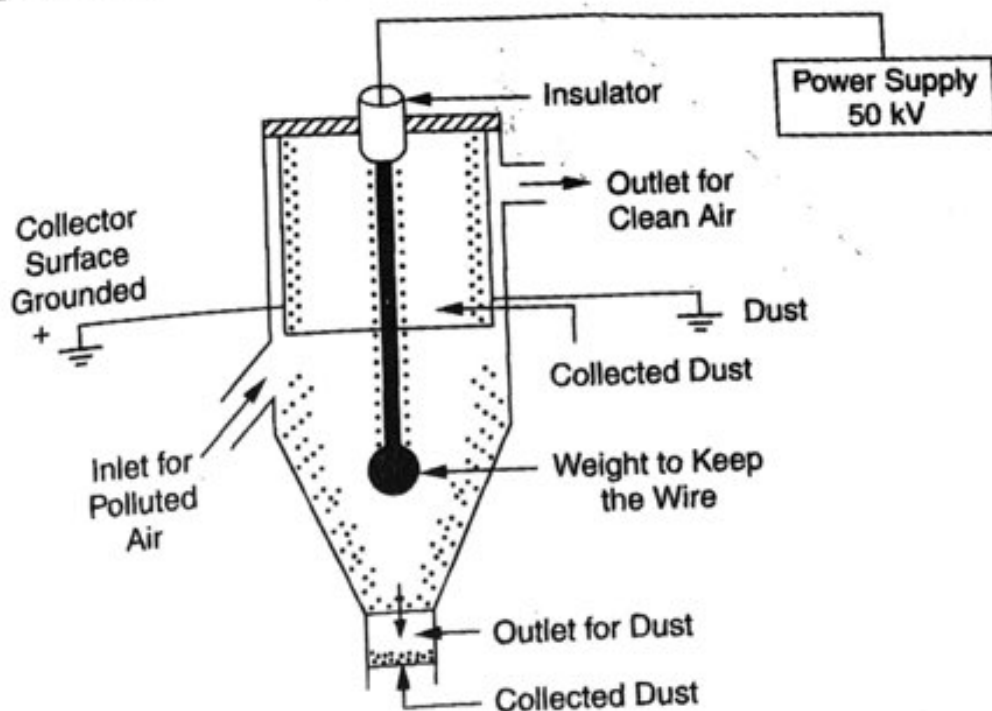


Fig. 3.13 Electrostatic Precipitator.

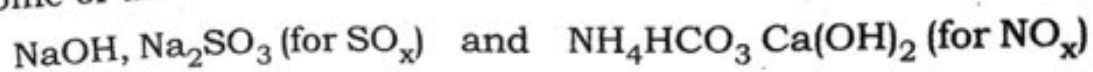
- (e) **Wet Scrubbers:** A natural cleansing mechanism is wet precipitation through which some pollutants are washed out of the atmosphere. The same concept is employed in wet scrubbers. The exhaust gases are passed through a fire spring of water so that the effective size of the particle is increased due to the formation of water particulate agglomerates. These are then separated from the gas stream by any of the collection mechanism.

However the collected waste water may become a source of water pollution.

- (ii) **Controlling Gaseous Pollutants:** Control of gaseous pollutants can be achieved through source correction methods. Some of the methods adopted for control of gaseous pollutants are:

- (a) **Combustion:** When the pollutant contains gases or vapours, which are organic in nature, flame combustion is employed. Combustion converts them to water vapours and CO_2 . Catalytic combustion is also employed where low operating temperatures are desired.

- (b) **Absorption/Adsorption:** Gaseous effluents are passed through scrubbers or absorbers. The technique can be dry or wet depending on the phase in which the main reactions occurs. Some of the solvents used for wet scrubbing are:



For dry scrubbing, activated carbon is used since it attracts thin layers of gases and vapours to its surface.

3.9 CARBON DIOXIDE SEQUESTRATION

Over the last century, human activity had a profound impact on the environment. Fossil fuel consumption, deforestation, and other unsustainable land use practices have resulted in a dramatic increase of carbon dioxide (CO_2) and other greenhouse gas (GHG) emissions into the atmosphere. Most scientists believe the increase of CO_2 emissions has created the human-induced climate warming conditions that are currently affecting the globe. If this trend continues, climate change will be the inevitable result. The long-term effects of global temperature change are largely unknown; however, adverse effects can already be seen in certain parts of the world in the form of droughts, increased severity of storms, and flooding, particularly in the poorer regions of the globe.

The natural production and absorption of carbon dioxide (CO_2) is achieved through the earth's biosphere and oceans. However, mankind has altered the natural carbon cycle by burning coal, oil, natural gas, and wood and each of these activities has increased in scale and distribution. Carbon dioxide was the first greenhouse gas demonstrated to be increasing in atmospheric concentration.

Atmospheric levels of CO_2 have risen well over 30% from pre-industrial levels of 280 parts per million (ppm) to present levels of 375 ppm. Evidence suggests this observed rise in atmospheric CO_2 levels is due primarily to expanding use of fossil fuels for energy. Predictions of global energy use in the next century suggest a continued increase in carbon emissions and rising concentrations of CO_2 in the atmosphere unless major changes are made in the way we produce and use energy - in particular, how we manage carbon. One way to manage carbon is to use energy more efficiently to reduce our need for a major energy and carbon source - fossil fuel combustion. Another way is to increase our use of low-carbon and carbon-free fuels and technologies (nuclear power and renewable sources such as solar energy, wind power, and biomass fuels). The most recent alternative for managing carbon is sequestration.

Carbon sequestration refers to the provision of long-term storage of carbon in the terrestrial biosphere, underground, or oceans, to reduce the buildup of carbon dioxide (the principal green house gas) concentration in the atmosphere. This is accomplished by maintaining or enhancing natural processes, or the development of new techniques to dispose of carbon.

The main natural sinks are (i) Oceans which take up on third of anthropogenic emissions of CO_2 , (ii) plants and other organisms that use photosynthesis to remove carbon from the atmosphere by incorporating it into biomass and release oxygen into the atmosphere, and (iii) soils which contain more organic carbon than is contained in vegetation and atmosphere together.

Enhancing Natural Sequestration

However, the natural CO_2 sequestration in the environment is not able to cope up with the increasing anthropogenic emissions, thus natural sequestration processes are enhanced. For instance, reforestation can increase the capacity of forests to act as sink. In the oceans, micrometer-sized iron particles in the form of hematite (iron oxide) or melanterite

(iron sulphate) are added to water to increase the carbon sequestration efficiency. The addition of iron particles stimulate the growth of phytoplankton, which in turn remove significant quantities of CO_2 from the atmosphere via photosynthesis. However, phytoplankton have a complex effect on cloud formation via the release of dimethyl sulfide (DMS) that is converted to sulfate aerosols in the atmosphere, providing cloud condensation nuclei (CCN). But the effect of small scale plankton blooms on overall production of DMS is still unknown. In soil, no-till farming, cover cropping and crop rotation methods are used to significantly enhance carbon sequestration. Conversion to pastureland with good management of grazing, can also increase carbon sequestration in soils.

Artificial Sequestration

For carbon to be sequestered artificially (i.e. not using the natural processes of the carbon cycle) it must first be captured, or it must be significantly delayed or prevented from being re-released in the atmosphere (by combustion decay, etc.) from an existing carbon-rich material, by being incorporated into an enduring usage (such as in construction). Thereafter it can be passively stored or remain productively utilized over time in a variety of ways.

In the oceans the carbon oxide is directly pumped into the water at depth and is expected to form lakes of liquid CO_2 at the bottom. However, this method could have dangerous consequences. The CO_2 could react with water to form carbonic acid, although most (as much as 99%) remain as dissolved molecular CO_2 .

The method of *geo-sequestration* or *geological storage* involves injecting carbon dioxide directly into underground geological formations. Declining oil fields, saline aquifers, and unminable coal seams have been suggested as storage sites. Caverns and old mines that are commonly used to store natural gas are not considered, because of a lack of storage safety.

Biosequestration of CO_2 involves plantation of biodiesel crops such as *Jatropha curcas*, switch grass and algal species etc.

According to Kyoto Protocol CO_2 should be brought down to 1990 level for the sustainable development. Thus CO_2 and other related green house gases are being considered for the carbon trading under *Clean Development Mechanism* (CDM) in the stock exchanges. CDM is an arrangement under Kyoto Protocol allowing industrialized countries with green house gas reduction commitment to invest in projects that reduce emissions in developing countries as an alternative to more expensive emission reduction in their own countries. An industrialised country that wishes to get credits from a CDM project must obtain the consent of the developing country hosting the project that it will contribute to sustainable development. The project is then submitted to CDM Executive Board for approval. Once the project is approved and registered, the CDM Executive board issues credits called *Certified Emission Reduction* (CERs) commonly know as *carbon credits* where each unit is equivalent to the reduction of one metric tonne of CO_2 or its equivalent) to the project participants based on the monitored difference between baseline and actual emissions. (Carbon credits are discussed in the last unit in detail).

3.10 WATER POLLUTION

When toxic substances enter lakes, streams, rivers, oceans and other water bodies, they get dissolved or lie suspended in water. This results in the pollution of water where by the quality of water deteriorates.

The word polluted water is defined as the deterioration of physical, chemical and biological properties of water brought about mainly by human activities and natural resources and which cause harmful effects on human and aquatic life.

Sources of Water Pollution: Water pollution is caused by natural as well as human activities.

- (i) Pollution from **natural sources** include the decomposed vegetable, animal and weathered products which are brought into main water resources.
- (ii) Pollution from **human activities** include domestic as well as industrial pollution.
 - (a) *Domestic:* The release of huge quantities of municipal and domestic wastes through drains into the rivers and canals is the major cause of pollution. The domestic waste water contain human faeces, kitchen wastes, organic water that provide nutrition to bacteria and fungi.
 - (b) *Industrial:* The industrial sources of water pollution include effluents from factories, refineries and a number of chemical industries. Water get polluted by acids, alkalies, detergents, copper, zinc, lead, mercury, pesticides etc. which are constantly added to water by industrial effluents, deteriorating the quality of water to such an extent that it becomes unsafe for human consumption.
 - (c) Apart from these two, the water is also polluted through **agricultural discharge** such as pesticides, insecticides plant, nutrients, fertilizers etc. Beside this, bacteria, algae, virus also cause water pollution.

3.11 CLASSIFICATION OF WATER POLLUTANTS

The problem of water pollution due to discharge of domestic and industrial water into water has already become a serious problem in the country. Nearly 75 to 80% of India's population is exposed, to unsafe drinking water as a result, enteric diseases often reach epidemic proportions.

To aid in a systematic discussion of water pollutants, they have been classified into nine categories as described below

1. Oxygen demanding wastes
2. Pathogens (Disease causing agents)
3. Synthetic organic compounds
4. Plant nutrients
5. Inorganic chemicals and minerals

6. Sediments
7. Radioactive substances
8. Thermal discharges
9. Oil

Water pollution is often caused by the combination of the above categories, increasing the severity of the problem. The categories are discussed briefly in the following sections:

1. Oxygen Demanding Wastes

Dissolved oxygen (DO) is essential for sustaining the plant and animal life in any aquatic system. Warm-water fish requires a minimum DO level of at least 5mg/L (5ppm). If DO level drops below the level necessary to sustain normal life, then water is classified as polluted.

The amount of DO in water is reduced because of oxygen demanding wastes. Oxygen demanding wastes are substances that oxidize in water, reducing the amount of DO. These substances include the organic substances which are formed due to death and decay of the algae and related species. Oxygen demanding wastes also include organic substances contained in municipal waste water or in effluents from certain industries, such as food processing and paper production. In addition, the oxidation of certain inorganic compounds may also contribute to the oxygen demand. Even naturally occurring organic matter, such as leaves and animal droppings, that find their way into surface water, add to the DO depletion.

The oxygen demanding wastes are oxidized by bacteria or micro organisms consuming dissolved oxygen in water to CO_2 and water. Thus DO levels drop. These substances produce undesirable odour, tastes and reduce the acceptability of water as a domestic supply.

There are several measures of oxygen demand commonly used.

- (a) Biological Oxygen Demand (BOD) BOD is the amount of oxygen required by microorganisms to biologically degrade the wastes. It is most important measure of the strength of organic pollution. It is discussed in more details in section —
- (b) Chemical Oxygen demand (COD) COD is the amount of oxygen needed to chemically oxidize the waste. It is discussed in more details in section —

2. Pathogens

Pathogens are disease causing organisms that grow and multiply within the host. Water is a potential carrier of pathogenic micro-organisms.. These pathogens are carried into the water bodies by sewage and wastes from farm and various industries, specially tanning and wheat packaging industries.

Contaminated water caused by poor sanitation can lead to both, water borne and water contact diseases.

Water borne diseases are those acquired by ingestion of pathogens not only in drinking water, but also from water that makes it into a person's mouth from washing food, utensils and hands.

Examples of pathogens associated with water borne diseases include bacteria, viruses and protozoa. A brief account of the diseases caused by these organisms given in table 3.7

Water-Contact diseases do not even require that individuals ingest the water. Just contact with contaminated water may cause disease. For example, Schistosomiasis (bilharzia) is the most common water-contact disease in the world, affecting approximately 200 million people, spread by free swimming larva in water, called cercaria. These larvae attach themselves to human skin, penetrate it and enter the blood stream.

According to WHO report approximately 80 percent of all sickness in the world is attributed to contaminated water.

Table 3.7: Diseases Commonly found in polluted water

Disease	Agent Type of organism	Symptoms
Cholera	Vibrio cholerae (Bacterium)	Severe diarrhea, vomiting, fluid loss, cramps and collapse.
Dysentery	Shigella dysenteriae (Bacterium)	Infection of the colon causes painful diarrhea with mucus and blood in the stools
Enteritis	Clostridium perfringens (Bacterium)	Inflammation of the small intestine causes general discomfort, loss of appetite, abdominal cramps and diarrhea
Typhoid	Salmonella typhi (Bacterium)	Early symptoms include headache, loss of energy, fever; later a pink rash appears along with hemorrhage in the intestine
Infectious Hepatitis	Hepatitis virus A (Virus)	Inflammation of liver causes jaundice, fever, headache, nausea, vomiting, severe loss of appetite;
Poliomyelitis	Poliovirus (Virus)	Early symptoms include sore throat, fever, diarrhea and aching in limbs. Spreads to spinal cord, paralysis and atrophy of muscles.
Amoebic Dysentery	Entamoeba histolytica (Amoeba)	Infection of the colon causes painful stools; abdominal pain
Schistosomiasis	Schistosoma (Fluke)	Disorder of the liver and bladder, causes blood in urine, diarrhea, weakness, lack of energy, repeated abdominal pain.
Ancylostomiasis	Ancylostoma (Hookworm)	Severe anemia, sometimes symptoms of bronchitis.

The coliform group of bacteria are considered reliable indicators of the hygienic quality of water. **Escherichia coli** (*E. coli*), belonging to the coliform group is harmless bacteria found in large number in human faeces. A large concentration of *E. coli* in water indicates faecal contamination and evidence of the presence of pathogens.

3. Synthetic Organic Compounds

These include pesticides, synthetic organic chemicals and detergents. In contrast to organic wastes, these compounds are not biodegradable and persist for longer periods. Most of these are accumulative toxic poisons and ultimately reach objectionable levels in water. Out of various synthetic organic compounds, the pesticides and detergents have been subjected to intensive study because of their liberal use resulting in water pollution.

Pesticides The term pesticide is used to cover a range of chemicals that kill organisms that humans consider undesirable. Pesticides can be classified into three main classes of insecticides, herbicides and fungicides. Other types include rodenticides, molluscicides and acaricides.

Pesticides enter the water body from surface run off from agricultural lands, waste discharge by pesticide manufacturers and by other means. Because of their world-wide usage, nearly all the rivers and oceans of the world contain pesticide residues.

There are three main groups under which the pesticides can be categorized — chlorinated hydrocarbons, organophosphorus compounds and carbonate pesticides. The common pesticides and their effects are tabulated in table 3.8

Table 3.8

Pesticide	Effects
1. Chlorinated hydrocarbons - DDT (Dichloro diphenyl trichloro ethane) Chlordane Heptachlor Aldrin Dieldrin Kepone Chlorophenoxy compounds (2,4,5,T, 2 - 4 D)	• Very persistent (last a long time in the environment) • Soluble in lipids and hence can accumulate in fatty tissues • Affect the calcium metabolism in predatory birds, resulting in birds, laying eggs with thin shells and consequent reproductive failure. • Tends to become more concentrated from one trophic level to another due to the process of *biological magnification. • Produce liver cancers • Cause birth defects • Severe neurological damage. • Contains the impurity dioxin, one of the most potent toxins known.

* Biological magnification implies that successively higher trophic levels in food chain are consuming food that has successively higher concentration of pesticides. For example, the concentration of DDT increases as we move from water → plankton → Fresh water fish → Predatory birds.

Pesticide	Effects
2. Organophosphorus compounds Parathion Malathion Diazinon	Non-persistent but damage nervous system. Acute exposure result in slurred speech, muscle twitching and convulsions.
3. Carbonate Pesticides Propoxur Carbaryl Aldicarb	Acute exposure results in nausea, vomiting, blurred vision and convulsions.

It may be emphasized that India is one of the few world's producers of DDT, which is banned in US in 1972 for environmental reasons.

Detergents: The term detergent means cleansing agent. The basic active ingredient in detergents is the surfactant or surface-active agent which contains hydrophobic (hydrocarbon) and hydrophilic (polar) groups.

Surfactants decrease the surface tension of water so that they can penetrate the surface and interstices of the object being cleaned. Although reducing the surface tension is an essential feature of cleaning agents, detergents usually contain 10 – 30 % surfactant. The remainder comprises of polysulphate salts (the builder) and a number of other ingredients. Both the surfactants and builders of detergents create serious pollution problems in water.

Surfactant concentrations as low as 1 ppm produce foam in rivers and in sewage treatment plants. Although these concentrations are non-toxic to humans, the presence of surfactants gives an off taste to drinking water. The presence of surfactants also reduce the rate of oxygen absorption in water and may lead to the symptoms similar to asphyxia in certain kind of fish (trout). Till the early 1960s the surfactants present in synthetic detergent was alkylbenzene sulphonate (ABS). ABS is non biodegradable and cause rivers and sewage treatment plants to become covered with huge amounts of foams. ABS have been now replaced by linear alkyl sulphonate (LAS) which is rapidly biodegradable.

The detergent builders pose greater problem to day. The polyphosphates builders are released into water and act as plant nutrients. The extensive growth of algae consumes most of dissolved oxygen from water.

Nitrilotriacetate (NTA) was considered a replacement but it proved to be hazardous to human health

Plant Nutrients (Sewage and agricultural runoff)

Nutrients are chemicals, such as nitrogen, phosphorus, carbon, sulphur, calcium, potassium, iron, manganese etc. that are essential to growth of living things. However, in terms of water, these nutrients are considered pollutants, when their concentrations are sufficient to allow excessive growth of aquatic plants, particularly algae. When these algae die and decompose they add undesirable odour and objectionable taste to water. Further, the decaying organic matter oxidizes and leads to reduced DO levels.

Pesticide	Effects
2. Organophosphorus compounds Parathion Malathion Diazinon	Non-persistent but damage nervous system. Acute exposure result in slurred speech, muscle twitching and convulsions.
3. Carbonate Pesticides Propoxur Carbaryl Aldicarb	Acute exposure results in nausea, vomiting, blurred vision and convulsions.

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The gradual accumulation of silt and organic matter in the lake is known as **eutrophication**. A eutrophic lake gets shallower and warmer with further growth and decay of algae and eventually lake turns into a marsh or bog.

The plant nutrients enter in the water through municipal waste water, industrial wastes and runoff from fertilized lands. The green slimy layer seen on the surface of some water bodies is an example of eutrophication of the lake. The slimy layer reduces light penetration and restricts atmospheric reoxygenation of water.

High concentration of nitrogen not only poses the problem of eutrophication but when found in drinking water, it is serious health hazard. Certain bacteria found in the intestine, reduce the nitrate in water to highly toxic nitrites (NO_2). Nitrites have a greater affinity for haemoglobin as compared to oxygen. Thus nitrite attaches to haemoglobin forming methaemoglobin. As a result, the oxygen carrying capacity of the blood is reduced producing a condition called methaemoglobin anemia commonly referred to *blue baby syndrome*. In young babies, it is extremely fatal. Further the nitrates can be converted into nitrosoamines in the body leading to gastric cancer.

5. Inorganic Chemicals and Minerals

This category of water pollutants includes inorganic salts, mineral acids, finely divided metals or metal compounds, trace elements, cyanides, organometallic compounds etc.

These pollutants are added to the water bodies through municipal and industrial waste waters and mine runoff. Most of these substances are toxic and are a health hazard to human life. Some of these pollutants are discussed below in brief.

Acid Mine Drainage: Mine drainage is an important source of increased acidity in natural water. Acid rain also adds to the acidity. The mining of sulphur bearing ores containing lead, zinc and copper lead to acid drainage. Coal mines discharge containing varying amounts of iron sulphide (pyrite) is also a major cause of acid mine drainage. These discharges release considerable quantities of sulphuric acid and ferric hydroxide, (which are formed as result of reactions between air, water and pyrite) into local streams through seepage. Acids cause corrosion of metals and concrete and is fatal to fish.

Soluble Salts: Water naturally accumulates variety of dissolved solids or salts as it passes through soils and rocks on its way to sea. These salts include cations such sodium, calcium, magnesium and potassium and anions such as chloride, sulphate and bicarbonate. The measure of salinity is to determine the concentration of total dissolved solids (TDS). Water with less than 500 mg/L TDS is considered safe for drinking purpose. For irrigation purposes, the TDS above 500 mg/L is also considered harmful.

Salts also cause 'hardness' of water leading to the formation of scale and sludges. High chloride concentration are fatal to fish. One way to approach salt problem is through desalination.

Heavy Metals: The term heavy metals' is less precisely defined. In chemical terms it refers to metals with specific gravity greater than about 4 or 5 metals. But more often, the term is simply used to denote metals that are toxic. The toxic properties of numerous metals have been known for years. The metals of particular concern in industrial waste waters are cadmium, chromium lead, mercury, silver, arsenic, aluminium, copper, cobalt, manganese etc. Some of these, such as chromium and iron, are extremely toxic in higher doses.

Metals may be inhaled or may be digested. These have a range of adverse impacts on the body, including nervous system and kidney damage, creation of mutations and induction of tumours. The kidneys contain millions of excretory units called nephrons and the chemicals that are toxic to kidneys are called nephrotoxins. Cadmium, lead and mercury are examples of nephrotoxic metals.

Table 3.9 summarises the effects of some of the toxic metals.

Table 3.9: Effects of trace metals present in air and water

Pollutant	Source	Effect
Lead	Automobile emissions, paints, storage batteries, lead piping, effluents from printing and dyeing industries	• Impairment of intellectual development and cognitive functioning • Absorbed through gastrointestinal and respiratory tract and cause liver and kidney damage. • Permissible limit in drinking water 0.05 mg/L
Cadmium	Effluents from textile, electroplating and chemical plants, Fertilizer industry cadmium nickel Satteries (Nicad)	• Inhalation of vapours causes kidney damage, bronchitis • Once absorbed by the organism remains resident for many years in the body. • High dose has been linked to lung cancers. • May produce bone defects. • Permissible limits in drinking water is 0.01mg/L
Chromium	Metallurgical and chemical industries, pigments for paints cement, paper, ceramic and glass industry.	• Low level exposure can irritate the skin and cause ulceration • Accumulates in the body of fish and adds to the health effects of eating the fish exposed to high levels of chromium. • Permissible limit in water 0.05mg/L

Pollutant	Source	Effect
Mercury	Various industrial effluents such as those from paint and paper manufacturing units. Highly prevalent in medical waste, lamps, thermometers, used in dentistry as amalgam for fillings.	- Builds up in the tissues of fish and other creatures living in contaminated water and may be carried up the food chain to humans. - It is proved to be the most toxic aquatic pollutant because of its rapid methylation in aquatic environment by some bacteria. Monomethyl mercury and dimethyl mercury and dimethyl mercury are highly toxic and cause neurotoxicological disorders.
Arsenic	Introduced as a by-product of fossil fuel combustion, irrigation water containing pesticides, chemical wastes	- Carcinogenic - Can cause mild bronchitis - Attack — SH group of enzymes and can coagulate proteins.
Silver	Electroplating and photographic industries	- Fish and lower organisms are susceptible to silver poisoning - It is cumulative poison and chronic ingestion can cause permanent discolouration. Permissible limit in drinking water is 0.05 mg/L.

6. Sediments

Sediments include soil, sand and mineral particles washed into the aquatic environment by storms and flood waters. Sediments are the sources of organic and inorganic matter in the streams, fresh water, rivers, and other water bodies. Soil particles eroded by running water ultimately find their way into water reservoirs and such a process is called 'siltation'. Reservoirs and dams are filled with soil particles and other solid materials because of siltation. This reduces their storage capacity. Further presence of suspended solid particles present in the water bodies block the sunlight and hence decrease the process of photosynthesis for aquatic plants. As a result, evolution of oxygen is decreased and it is not enough as required by aquatic animals.

- Sediments increase the cost of water treatment used for culinary purposes.
- Sediments enter pumping equipment and power turbines and increase the turbidity.
- In suspension, the solids may cause thickening of fish gills which may lead to eventual asphyxiation of the fish.

7. Radioactive Substances

Radioactive pollutants enter into the aquatic system through a number of human activities involving the use of naturally occurring or artificially pro-

duced radioactive materials. The radioactive pollutants enter into the water system through a variety of sources such as

- Mining and processing of ores to produce radioactive ores, e.g. uranium and thorium.
- From nuclear power plants and from industrial use of radioactive materials
- Leakage from underground nuclear detonations
- Use of radio isotopes in medicine, industry, agriculture and research operations

The refining of uranium ore is an important source of radioactive waste, producing radionuclides of radium, bismuth etc. Radium is considered to be a hazard in drinking water. It causes various genetic disasters in human beings. Radioactive substances can enter humans with food and water and get accumulated in blood and certain vital organs like thyroid gland, the liver, bone and muscular tissues. These may cause cancers, leukemia and eye cataract.

8. Thermal Discharges

Industry and power plants use large quantities of water for cooling purposes. Used coolant water is usually discharged directly into water bodies. This results in increase in temperature of the water bodies which is called thermal pollution. Coal fired power plants, electric powers, steel and chemical industries as well as atomic energy plants discharge their heated effluents into nearby lakes or rivers leading to thermal pollution.

Rise in temperature of water decreases dissolved oxygen content of water which affects the aquatic life. At 32°F the DO content of water is 15 ppm which is decreased to 6.5 ppm at 64° F. Fishes are killed due to the action of heat on nervous system, inactivation of enzymes and coagulation of cell protoplasm.

An increase in temperature also increases the toxicity of some chemical pollutants.

9. Oil

Oil is an important commodity used in one way or the other. Oil and oil wastes are added to the water bodies from industries as effluents, oil refineries, storage tanks, automobile waste oil and petrochemical plants. Oil is insoluble in water and hence floats over it as a thin layer. This oil may penetrate the feathers of the birds and affect their insulation and buoyancy. Thus birds experience difficulty in floating and flying. The birds may ingest oil while they dive to feed. This ingestion may produce toxic effects.

Moreover the oil slick formed on the surface prevents the diffusion of oxygen into water resulting in decreased concentration of DO.

10. Volatile Organic Compounds

These are one of the most commonly found contaminants in water. Volatile organic compounds are often used as solvents in industrial processes.

They are volatile and hence their concentrations remain as low as few micrograms per litre in surface water. But in ground water their concentrations can be hundreds or thousands of times higher. Volatile organic compounds are toxic and their presence in drinking water is cause of great concern. The following VOCs are tabulated in table 3.10

Table 3.10: Volatile Organic Compounds

Vinyl Chloride	Production of polyvinyl chloride resins	Carcinogenic
Trichloroethylene	Used as a solvent, heat transfer medium, manufacture of CFCs.	Causes tumours in animals.
Trichloroethylene	Solvent, used to clean electronic parts	Carcinogenic
1,2 Dichloroethane	Metal degreaser, used in manufacture of vinyl chloride, varnish removers and soap compounds.	Cause injury to central nervous system liver and kidneys
Carbon tetrachloride	Common household cleaning agent, used in fire extinguisher, solvent	very toxic, only a few millilitres can result in to death

3.12 WASTE WATER TREATMENT

Until 1900's human wastes were simply dumped as raw sewage into the nearest stream or river. But later on when a number of diseases were linked to the waste waters, the treatment of waste water was taken up seriously.

The purpose of waste water treatment is to remove the contaminants from water so that the treated water can meet the acceptable quality standards. The quality standards usually depend upon whether water will be reused or discharged in to a receiving stream..

The waste water is often referred to as *sewage*. Sewage is the liquid water which includes human and domestic waste waters, industrial wastes, ground wastes and street washings. Sewage contains 99.9 percent water and rest being the organic, inorganic compounds in dissolved colloidal or suspended states. It is this small percentage of pollutants that brings about polluted conditions.

Constituents of Sewage

There are various types of chemicals founds in waste water, depending upon the source from where it is being discharged. Depending upon the nature of the sewage it can be grouped as

- Domestic sewage:* Domestic sewage includes human excreta as well as discharges from kitchens, baths, lavatories etc. from public and private buildings.

- (ii) **Industrial Sewage:** Industrial sewage or waste water may be regarded as the liquid waste water in which industrial effluents are present as main wastes
- (iii) **Storm water:** This is the rain water which flows as runoff from streets, open yards etc.
- (iv) **Combined Sewage:** Combined sewage is the combination of domestic sewage, industrial waste and storm water.

Characterisation of Waste Water (Strength of the Sewage)

The strength or impurity in a particular sewage is usually measured on the following categories.

1. Dissolved Oxygen

Oxygen is poorly soluble in water. The amount of oxygen in water depends on physical, chemical and biological activities taking place in water. The solubility of dissolved oxygen in water at saturation at any temperature and pressure is given by Henry's law.

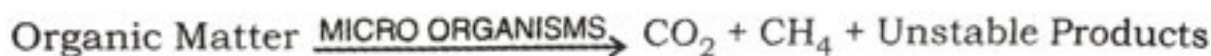
For determination of dissolved oxygen refer to chapter 1.

Biochemical Oxygen Demand

When biodegradable organic matter is released into water, the micro organisms feed on the wastes and break it down into simpler organic or inorganic substances. If the decomposition takes place in the **presence of oxygen** i.e aerobically, **the non objectionable**, stable end products are formed.



On the other hand if **insufficient oxygen** is available, the decomposition takes place anaerobically. The micro organisms causing the decomposition of the organic matter without the presence of oxygen are entirely different from aerobic bacteria and produce **highly objectionable** end products including H_2S , NH_3 and CH_4 .



The amount of oxygen required by micro-organisms to oxidize organic wastes aerobically is called biochemical oxygen demand (BOD). The BOD of raw water will indicate the extent of organic matter present, thus indicating the extent of treatment required to make it safe for use. The BOD of treated water should be nil, so as to make it free from any organic matter. If BOD is high, the dissolved oxygen becomes low and this results in greater pollution.

Determination: Polluted waters will continue to absorb oxygen for many months, till the oxidation gets completed and hence it is not practical to determine this ultimate oxygen demand which would require an extended

period of time. As a result, it has become standard practice simply to measure the oxygen over a shorter period of 5 days. This is known as 5-day BOD test. **BOD₅ is the total amount of oxygen consumed by microorganisms during the first five days of biodegradation.**

In its simplest form BOD₅ would involve diluting a known volume of sample water with a known volume of pure water, whose oxygen content is already known. This is kept for 5 days at 20°C in a stoppered bottle, away from light. The dissolved oxygen is measured after the period of incubation. The difference between the original oxygen content and residual oxygen content will indicate the oxygen consumed by the water sample in five days in oxidizing the organic matter present in water sample. Thus BOD₅ is given by:

$$\text{BOD}_5 = \text{Loss of oxygen in mg/L} \times \text{Dilution factor}$$

Chemical Oxygen Demand

Although BOD test is applicable to organic wastes, there are many drawbacks. Certain organic materials are not biodegradable and hence can give wrong conclusions that less organic matter is present because BOD of such water samples will be low. In this case COD, Chemical Oxygen demand reveals the real organic content present. Here the oxidation of organic substances present in water is done chemically.

COD, is the amount of oxygen required by organic matter in a sample of water for its oxidation by a strong chemical oxidizing agent such as $\text{K}_2\text{Cr}_2\text{O}_7$.

Determination: A known amount of $\text{K}_2\text{Cr}_2\text{O}_7$ is added to a measured amount of the sample and the mixture is boiled with concentrated H_2SO_4 . After boiling, the amount of unreacted $\text{K}_2\text{Cr}_2\text{O}_7$ is determined by titrating against a standard Mohr's salt solution using ferroin as indicator.

The difference between the dichromate originally present and the dichromate remaining unreacted gives the amount of dichromate used for oxidation of organic matter.

2. Dissolved CO₂

Varying amounts of CO₂ are present in water. CO₂ can be picked up from the atmosphere or it is generated at the bottom of the lake due to the decay of organic matter. Rivers receiving acid wastes may also show high CO₂ content.

Determination: CO₂ is determined by titrating the sample water with N/10 sodium carbonate solution using phenolphthalein as indicator. The end point is appearance of pink colour.

3. Free Chlorine

Chlorine determination is necessary in those samples of water that have been treated with chlorine or hypochlorites.

Determination: The estimation of free chlorine is based on oxidation of KI by free chlorine i.e. by iodometric titration. (Refer to chapter 1 for details)

4. Dissolved Chlorides

The presence of chlorides in natural waters is due to the discharges of effluents from the industries, irrigation drainage, presence of marine sedimentary deposits etc.

Determination: Chlorides ions can be determined by titration of the water sample against a standard solution of AgNO_3 using potassium chromate as indicator. (Argentometric titration). (Refer to chapter 1 for details.)

5. Total Dissolved Solids (TDS)

Dissolved solids denote mainly the various kinds of minerals present in water. However, some organic substances present in water also contribute to the dissolved solids.

Determination: The amount of total solids in a water sample can be determined very easily as the residue left after evaporation of the filtered sample. Refer to chapter 1 for details.

3.13 WASTE WATER TREATMENT PROCESSES

Available waste water treatment processes can be broadly classified as

- Physical processes
- Chemical processes
- Biological processes

These processes consists of a series of unit operations. These are applied in different combinations composition and specifications of the waste water.

Further, the waste waters treatment processes are generally grouped according to the water quality they are expected to produce. These processes are (physical, chemical and biological) grouped as

- (i) *Primary Treatment:* Removes suspended solids and floating matter
- (ii) *Secondary Treatment:* Involves biological process and organic matter is removed.
- (iii) *Tertiary or Advanced treatment:* Physical, chemical or biological processes depending upon the impurities to be removed.

We would be discussing these in detail in the following section.

1. Primary Treatment

Primary treatment plant utilizes physical processes such as screening or sedimentation to remove a portion of pollutants that will settle, float or that are too large to pass through simple screening devices.

Primary treatment comprises two steps, Pretreatment and Sedimentation.

- (a) *Pretreatment:* Pretreatment consists of screening and grit removal. Screening removes large floating objects such as rags, sticks, wood and other large floating and suspended solids. A typical screen consists of a parallel steel bars spaced anywhere from 2 to 7 cm apart.

This may be followed by a wire mesh screen. The material collected on the screens is disposed off. One way to dispose off the material is to use a device called comminuter, which grinds the coarse material in to small pieces which flow along with water and handled in sedimentation tank. After screening, the waste water passes into a grit chamber where the velocity of water is reduced or it is detained for a few minutes. Here sand, grit and other heavy materials settle down. The grit settling chambers are periodically disconnected from the main system to remove grit manually, for possible use in land-filling, road making and on sludge drying beds. Grit also is a good manure for growing crops.

Sedimentation: From the grit chamber, the sewage passes to a primary settling tank known as sedimentation basin. Here the velocity of the water is reduced considerably to allow most of the suspended solids to settle out by gravity. The most common equipment used include horizontal flow sedimentation tanks, similar to the one described under domestic water. The water is detained in the horizontal flow tanks for 2-3 hours resulting in removal of 50% of the suspended solid matter. An efficient sedimentation tank or clarifier removes about 80-90% of the suspended solids and 40% of organic matter. The solids that settle are called primary sludge or raw sludge.

Sedimentation aids: The removal of finely divided solids is difficult by simple sedimentation technique. In such cases, mechanical flocculation chemical coagulation is employed.

In *mechanical flocculation*, the waste water is passed through a sedimentation tank which is fitted with rotating paddles which rotate slowly at speed of 0.43 m/s. This slow mechanical stirring allows the finely divided solid particles to coalesce into larger particles and settle out.

In *chemical coagulation*, certain chemical substances, known as coagulants are added. Coagulation used for sewage treatment are similar to the ones used in water treatment. The coagulants react with colloidal matter in the sewage to form floc. The floc entraps the smaller particles and eventually settles down as sludge. Common coagulants used in sewage treatment are:

Alum $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$, Copperas $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, hydrated lime, ferric chloride, and chlorinated copperas, $\text{FeSO}_4 \cdot \text{Cl}$ (mixture of ferric sulphate and chloride).

Apart from using coagulants, coagulant aids are also used sometimes for efficient removal of finely divided particles. The commonly used coagulant aids are, activated silica polyelectrolytes (polymers of cyamide or methacrylic acids), and hydrolysed high molecular weight polymers. The coagulant aids draw a number of suspended particles together because of their long chain molecular structure. They also neutralize the charge on the particles by them bringing them closure to form large particles.

In *industrial waste water*, sometimes the primary treatment also includes equalization and neutralization

Equalization: Sometimes different types of wastes are produced by some industries. To apply uniform treatment, different effluents are held in big

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In *industrial waste water*, sometimes the primary treatment also includes equalization and neutralization

Equalization: Sometimes different types of wastes are produced by some industries. To apply uniform treatment, different effluents are held in big

tanks for certain periods and are mixed thoroughly to produce homogeneous equalized effluent.

Neutralization: Acidic wastes are neutralized with lime stone and alkaline wastes are neutralized by treatment with sulphuric acid or CO_2 or waste boiler flue gas. If both acid and alkaline wastes are produced in the nearby plants then mutual neutralization by mixing there is the cheapest method of neutralization.

2. Secondary Treatment

In primary treatment, only those materials are taken care of, which can be removed by simple physical or mechanical action. However, in waste water much of the organic material is dissolved or in colloidal form which is not removed by primary treatment. This organic material is oxygen demanding and must be removed further. Thus the main purpose of the secondary treatment is to remove the organic matter. This is achieved through biological processes which are quite similar in concept to the natural biodegradation of organic matter by aerobic bacteria. The various changes which are brought by secondary treatment include:

- Coagulation of the finely divided or colloidal matter.
- Oxidation of the organic matter to CO_2
- Conversion of nitrogenous organic matter to ammonia, which is eventually converted into nitrite and nitrate.
- Anaerobic digestion of the sludge so obtained.

There are three commonly used approaches, all of which take advantage of the biological processes. These processes are

- (i) Trickling filters.
- (ii) Activated Sludge Process
- (iii) Oxidation Ponds (Lagoons).

(i) Trickling Filters (Aerobic filtration)

A trickling filter consists of a rotating distribution arm that sprays the liquid over a circular bed of rocks or other coarse material (Fig. 3.14)

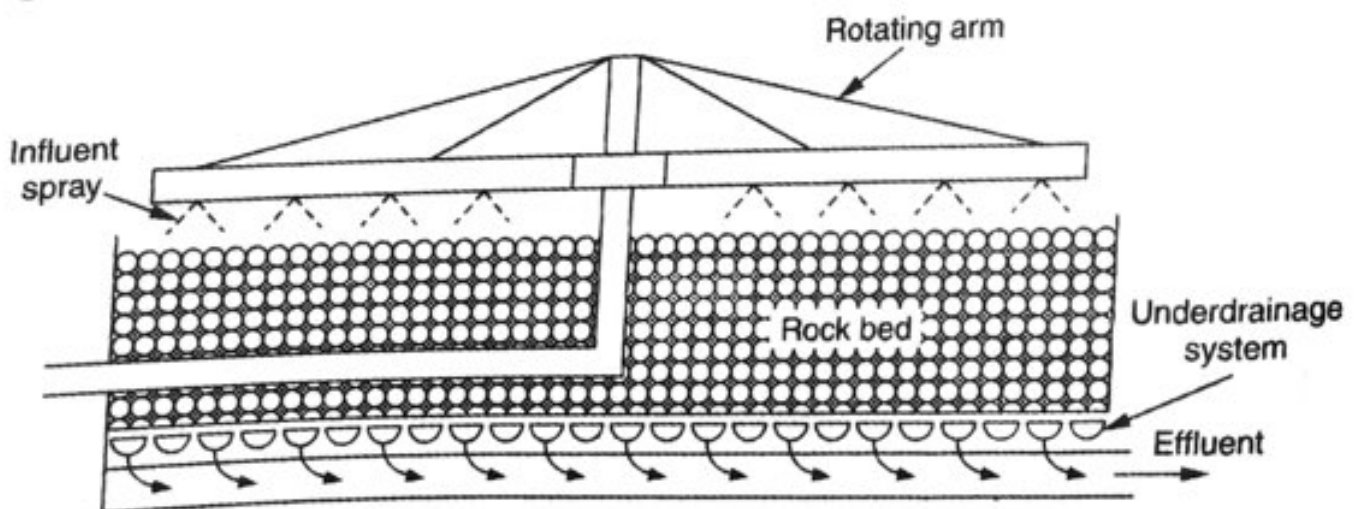


Fig. 3.14 Sketch of the cross section of a trickling filter.

The rock bed is 1 to 3 meters deep with enough open spaces between rocks to allow air to circulate easily. The individual rocks become coated with a layer of biological slime or a gelatinous film that absorbs and consumes the wastes trickling through the bed. The slime consists mainly of aerobic microorganisms (bacteria, algae, protozoa etc) which are collectively called **zooglea** and thrive on the nutrients supplied by the waste water. The influent (waste water) is sprinkled over this bed packing. As the liquid trickles over the packing, oxygen and the dissolved organic matter diffuses into the slimy layer and are oxidised by the microorganisms to form CO_2 , NO_3 , etc., which diffuse back out of film and appear in the filter effluent. As the microorganisms utilize the organic matter, the thickness of the slime layer increases and it gets detached from the surface of the rock and is carried away along with the effluent, which is referred to as *Humus*. The effluent along with detached slimy layer and treated water is collected at the bottom of the filter and passed on to settling tank, where the detached slimy layer settles down and is removed. The slimy layer (sludge) is then pumped to the sludge digestion unit.

More recently, plastic filters have been used for trickling filters. The plastic filters are light and can be stacked many times higher than the conventional bed rock which crumbles under its own weight if stacked high. These filters, made with the plastic media are sometimes referred to as 'biological towers'.

Advantages of the Process

- Trickling filters are simple to operate and can produce BOD removal to the extent of 65 to 85%
- Constant monitoring is not required.
- Effluents so produced are of better quality.

Limitations

- The microbial film formed is sensitive to temperature changes in winters.
- Efficiency of the filter is dependent upon the composition of waste, pH, size uniformity of the filtering medium and the supply of air.
- Cost of construction is high
- Trickling filters are used for treating industrial waste water from dairy, brewery, food processing, pulp and paper mills, pharmaceuticals, petrochemicals etc.

(ii) Activated Sludge Process

This is the most versatile biological oxidation method, employed for the treatment of waste water containing organic matter. It is a biological waste water treatment process in which a mixture of waste water and activated sludge is agitated and aerated. *The activated sludge is the sludge obtained by settling the sewage in presence of excess of oxygen.* The activated sludge is biologically active because it is heavily laden with microorganisms which are in active state of growth.

The activated sludge treatment process, the effluent after the primary treatment, is passed to an aeration tank where the activated sludge obtained from the previous process is present. The organic matter in the waste water is brought into intimate contact with the activated sludge in presence of adequate amount of oxygen or air. During this aeration process, oxidation of organic matter present in waste water takes place. The oxidation is carried out by microorganisms which are aggregated in masses about .1mm in diameter. To maintain aerobic conditions, the air or oxygen is pumped into the tank and the mixture is thoroughly agitated. After about 6-8 hours, the waste water, now referred to as 'mixed liquor' is made to flow through a settling tank where the solid, mostly bacterial masses are separated from the liquid. This settled mass, called the sludge, is partly reintroduced in the aeration tank to provide an effective microbial population for a fresh treatment cycle and is therefore termed as activated sludge. Rest of the sludge is digested in a sludge digester, along with the primary sludge obtained from primary treatment. Fig. 3.15.

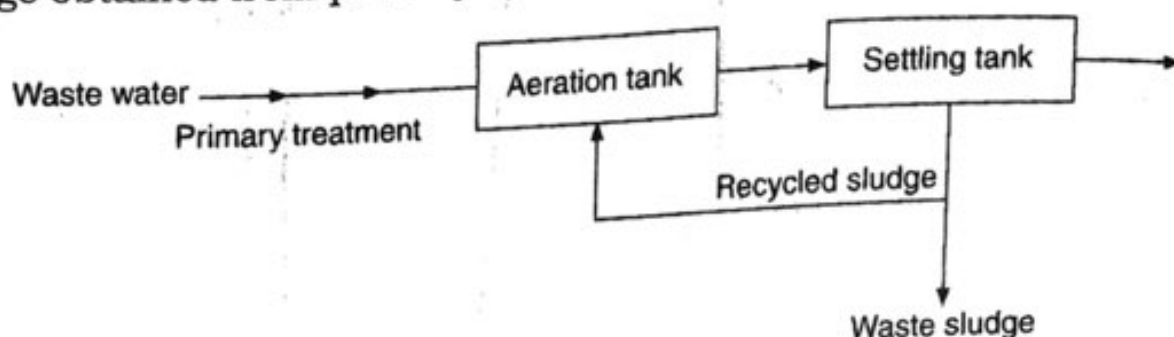


Fig. 3.15 Activated sludge process

Advantages

- The primary advantage of the activated sludge process is, good effluent quality. The effluent after going through activated sludge has little BOD ($< 20\text{mg/L}$)
- It takes less area as compared to trickling water filters.
- The activated sludge process equipment is less expensive.

Limitation

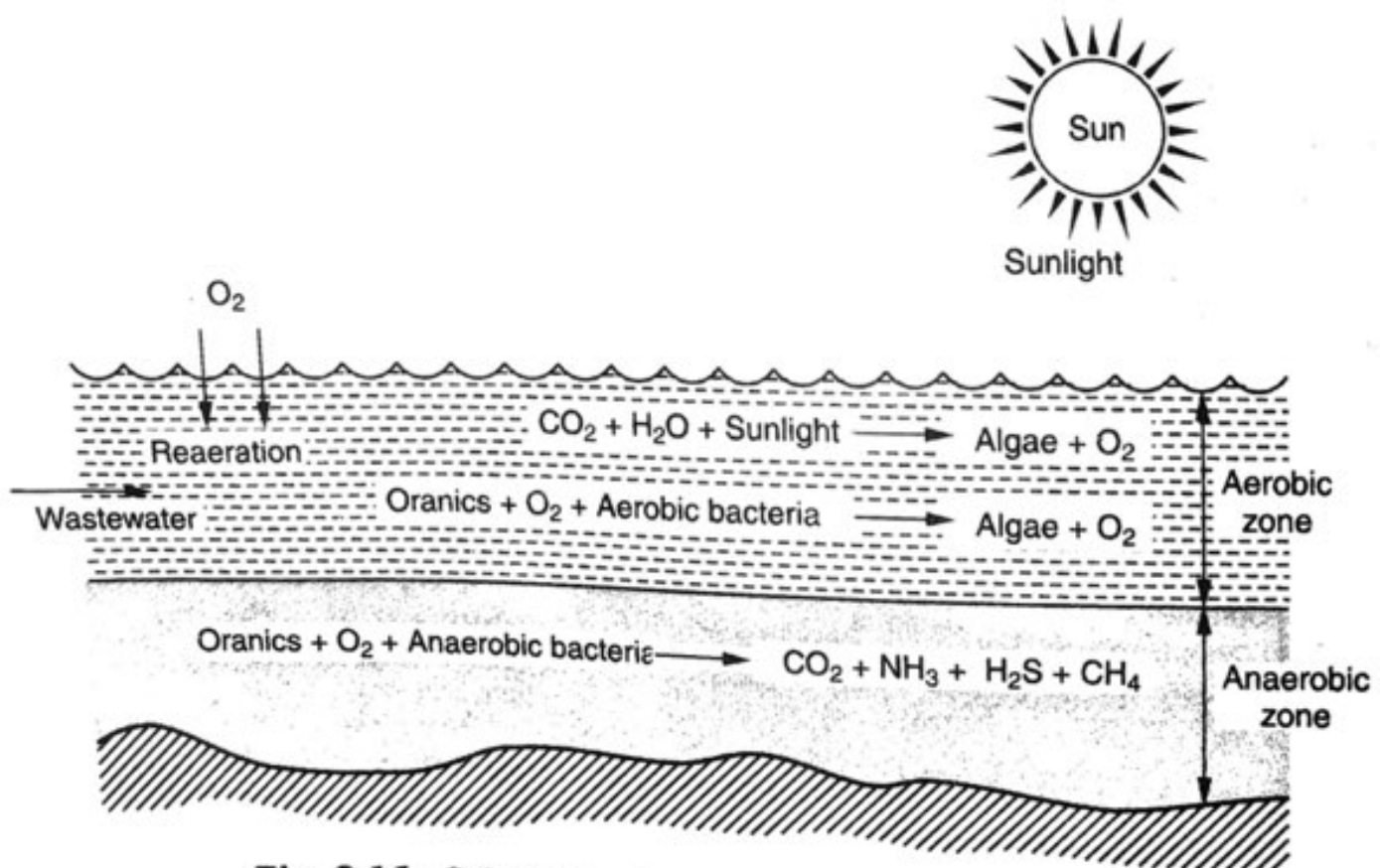
- For the process to be efficient, at least 0.5 ppm oxygen must be present all time.
- The optimum pH range 6.5 to 9.0 has to be maintained throughout.
- Low temperature slows down the rate of metabolism, while high temperature increases the metabolic activity to such an extent that the oxygen is consumed fast, leading to anaerobic conditions.
- The presence of detergents (which are not biodegradable) lead to the formation of foam, making the process difficult.
- The disadvantage of activated sludge process is the production of a huge amount of sludge, which should be digested and disposed off.
- Table 3.11 gives a comparison between trickling filters and activated sludge process for water treatment.

Table 3.11: Comparison of trickling filters with activated sludge systems

Trickling filters	Activated sludge systems
1. Bacterial growth is fixed on the media	1. Bacterial growth is suspended as a dispersed floc
2. All solids from the settler are wasted	2. Solids from the settler are partially recycled
3. Less sensitive to shock loading, more stable	3. More sensitive to shock loadings, required closer process control
4. Produce odours	4. Produce spray clouds
5. Less effective in removing disease-causing organisms	5. More effective in removing pathogens than trickling filters
6. Low operating costs	6. High operating costs

(iii) Oxidation Ponds

Oxidation ponds are large shallow ponds, typically 1-2 m deep, where the partially treated waste water is decomposed by microorganisms. The waste water enter the pond at one end and of the treated waste water is collected at the other end. The organic matter is oxidized by the microorganisms present in the pond (such as pseudomonas, flavo bacterium etc). The decomposition of the organic matter near the surface is aerobic, whereas near the bottom, it is anaerobic. Such ponds having a mix of aerobic and anaerobic conditions are called facultative ponds. The oxygen required for aerobic decomposition comes from algal photosynthesis. However deeper ponds known as lagoons are mechanically aerated. Fig. 3.16

**Fig. 3.16** Schematic diagram of an oxidation pond.

Advantages

- The process is simple and cheap.
- Can be used for all types of waste waters
- Due to the light pH of waste water in the pond, the heavy metal ions present in waste water are precipitated as hydroxides which settle as sludge.

Limitations

- The oxidation ponds require larger space.
- Anaerobic conditions may lead to release of bad odours.

The main drawback of the above secondary treatment processes is the formation of sludge. The collection, processing and disposal of sludge can be the most costly and complex aspect of waste water treatment.

Sludge Treatment and Disposal

Sludge is the watery residue from the primary sedimentation tank and humus tank from secondary treatment. The quantity of sludge produced may be as high as 2 percent of the original volume of waste water, depending upon the treatment process used. One of the primary goals of sludge treatment is to separate as much of the water from the solids as possible. In addition to reducing the water content of the sludge, it may be stabilized so that its biological activity and tendency towards putrefaction is reduced.

The traditional method of sludge digestion is **anaerobic digestion**. It involves the microorganisms that thrive in absence of oxygen. The organic material in the sludge is digested by these microorganisms under anaerobic conditions to give carbon dioxide and methane gas. The components of the sewage which can be converted into gases are called volatile solids. Though the sludge digestion is complex, it can be summarized in two steps.

- First the complex organics such as fats, proteins, and carbohydrates are converted to organic fatty acids by acid forming bacteria.
- Then these acids are converted into methane, CO_2 etc by methane forming bacteria.

Most treatment plants use a sludge digester in which the sludge is maintained at 35° for 30 days at a pH of 7.0 to 8.0. CH_4 , CO_2 and NH_3 are liberated as the end products (Fig. 3.17)

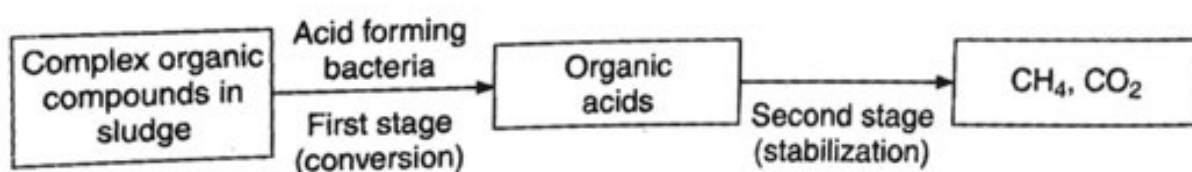


Fig. 3.17 Anaerobic digestion

Digested sludge is removed from the anaerobic digester. This sludge contains 90 to 93% water and is *dewatered*. Dewatering is accomplished by mechanical methods, the most common being centrifugation and filtration,

which includes pressure filtration and vacuum filtration. Drying beds are also commonly used. The dewatered sludge is sent for ultimate disposal. Wet sludge is sprayed on to crop land where it functions as fertilizer. Dried sludge may be used as a landfill or a soil conditioner.

Tertiary Treatment

Any treatment that follows primary and secondary treatment is considered tertiary treatment. Tertiary treatment improves the quality of the effluent further.

The effluent after secondary treatment plant still contains suspended solids (20-40mg/L) which may settle on the stream or river bed and inhibit certain forms of aquatic life. Some amount of BOD, significant amount of nutrients, dissolved solids, traces of organic chemicals and other contaminants are also present. The emphasis on recovery of valuables from industrial wastewaters have created the need for tertiary treatment.

The type of tertiary treatment depends upon the specific goal which include removal of

- suspended solids
- bacteria
- dissolved organic solids
- toxic substances
- nutrients (phosphorus and nitrogen)

Depending upon the required quality of final effluent a number of techniques can be used. Some of which are described in brief as follows:

(i) **Removal of suspended solids (Microstraining):** This can be achieved by **microstraining**. The filter media consists of finely woven stainless steel fabric. The treated waste is allowed to pass through it. The solids which are retained on the fabric are washed into a trough, which recycles the solids to the sedimentation tank.

(ii) **Removal of dissolved solids:**

(a) *Adsorption:* Dissolved solids can be organics or inorganic. The soluble organics are removed by **adsorption** on activated carbon. Special adsorbents are commercially available for the removal of toxic heavy metals from industrial waste water.

(b) *Solvent Extraction:* It is sometimes used to recover phenolic materials from waste waters of refineries and coke plants. The waste water is intimately brought in contact with a solvent having high affinity for the solute. The layers (aqueous and the solvent layer containing solute).

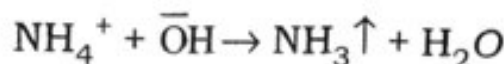
(c) *Ion Exchange:* Ion exchange is extensively used to remove hardness and iron and manganese salts from drinking water as described in chapter 1. The same technique has been extended to waste water treatment for the removal and recovery of waste water treatment.

- (d) **Reverse Osmosis:** When waste water containing dissolved solids is allowed to pass through a semi-permeable membrane at a pressure, which is more than osmotic pressure, the water from the waste passes through the membrane. Hence a highly concentrated solution containing dissolved solids is left behind.
- (e) **Chemical precipitating:** The precipitating agents like lime etc remove heavy ions by precipitating these as hydroxides. Precipitating agents include FeSO_4 , alum and ferric chloride.

(iii) Removal of Nutrients

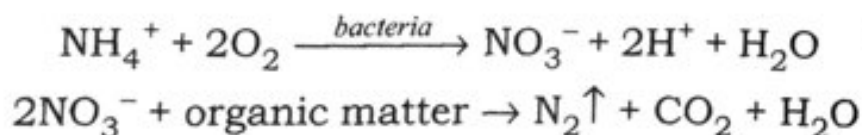
- (a) **Nitrogen Removal:** All forms of nitrogen in wastewater are harmful because plants can utilize the inorganic forms as nutrients, NH_3 can be utilized by bacteria resulting in reduced oxygen in water.

One of the method of nitrogen removal is **ammonia stripping**. Ammonia is present in natural water as ammonium, in (NH_4^+) . This NH_4^+ is changed to ammonia gas by raising the pH (the OH^- concentration) of the waste water by adding quick lime.

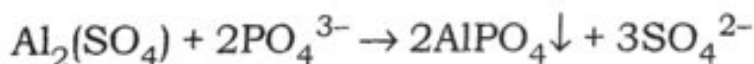


The ammonia gas is liberated

Another approach of nitrogen removal is nitrification i.e. to convert NH_4^+ to NO_3^- , followed by anaerobic stage in which microorganisms convert nitrates to nitrogen gas (N_2)



- (b) **Phosphorus Removal (Chemical precipitation):** Phosphorus is present in the form of orthophosphates (H_2PO_4^- , HPO_4^{2-} and PO_4^{3-}) Phosphates are removed by adding coagulants usually alum [$\text{Al}_2(\text{SO}_4)_3$] or lime [$\text{Ca}(\text{OH})_2$].



- (iv) **Removal of bacteria:** Bacteria are removed by retaining the effluents in maturation ponds or lagoons for specified period of times.

3.14 WASTE WATERS FROM SOME TYPICAL INDUSTRIES

1. Textile Industries

Introduction

The textile industry uses vegetable fibers such as cotton; animal fibers such as wool and silk; and a wide range of synthetic materials such as nylon, polyester, and acrylics.

The stages of textile production are fiber production, fiber processing and spinning, yarn preparation, fabric production, bleaching, dyeing and

printing, and finishing. Each stage generates wastes that require proper management.

Waste Characteristics

- Textile production involves a number of wet processes that may use solvent.
- Emissions of volatile organic compounds (VOCs) mainly arise from textile finishing, drying processes, and solvent use.
- Process wastewater is a major source of pollutants. It is typically alkaline and has high BOD — from 700 to 2,000 milligrams per liter (mg/l) — and high chemical oxygen demand (COD) in the range 2 to 5 times BOD.
- Wastewater also contains solids, oil, and possibly toxic organics, including phenols from dyeing and finishing and halogenated organics from processes such as bleaching.
- Dye wastewaters are frequently highly coloured and may contain heavy metals such as copper and chromium.
- Wool processing may release bacteria and other pathogens.
- Pesticides are sometimes used for the preservation of natural fibers, and these are transferred to wastewater during washing operations.
- Pesticides are used for mothproofing, brominated flame retardants are used for synthetic fabrics, and isocyanates are used for lamination.
- Air emissions include dust, oil mists, acid vapours, odours, and boiler exhausts.
- Cleaning and production changes result in sludges from tanks and spent process chemicals, which may contain toxic organics and metals.

Treatment Technologies

- VOC abatement measures include using scrubbers, employing activated carbon adsorbers, and routing the vapors through a combustion system.
- A common approach to wastewater treatment consists of screening, flow equalization, and settling to remove suspended solids, followed by biological treatment.
- Physical-chemical treatment is also practiced: careful control of pH, followed by the addition of a coagulant such as alum before settling, can achieve good first-stage treatment. Further treatment to reduce BOD, if required, can be carried out using oxidation ponds or another aerobic process; up to 95% removal of BOD can be achieved.
- Anaerobic treatment systems are not widely used for textile wastes.
- Carbon adsorption is sometimes used to enhance removal. In some case, precipitation and filtration may also be required.

- Residues and sludges often contain toxic organic chemicals and metals. These should be properly managed, with final disposal in an approved, secure landfill.
- Sludges containing halogenated organics and other toxic organics should be effectively treated by, for example, incineration before disposal of the residue in a secure landfill.

2. Tanning and Leather Finishing Industries

Introduction

In the tanning process, animal hides and skins are treated to remove hair and nonstructured proteins and fats, leaving an essentially pure collagen matrix. The hides are then preserved by impregnation with tanning agents. Leather production usually involves three distinct phases: preparation (in the beamhouse); tanning (in the tanyard); and finishing, including dyeing and surface treatment. A wide range of processes and chemicals, including chrome salts, used in the tanning and finishing processes.

The tanning and finishing process generally consists of :

- Soaking and washing to remove salt, restore the moisture content of the hides, and remove any foreign material such as dirt and manure
- Liming to open up the collagen structure by removing interstitial material
- Fleshing to remove excess tissue from the interior of the hide
- Dehairing or dewooling to remove hair or wool by mechanical or chemical means
- Bating and pickling to delime the skins and condition the hides to receive the tanning agents.
- Tanning to stabilize the hide material and impart basic properties of the hides
- Retanning, dyeing, and fat-liquoring to impart special properties to the leather, increase penetration of tanning solution, replenish oils in the hides, and impart color to the leather
- Finishing to attain final product specifications.

Waste Characteristics

- Composite untreated wastewater, amounting to 20—80 cubic meters per metric ton (m^3/t) of hide or skin, is turbid, clouted, and foul smelling. It consists of acidic and alkaline liquors, with chromium levels of 100—400 milligrams per liter (mg/l) sulfide levels of 200—800mg/l – nitrogen levels of 200—1,000mg/l – biochemical oxygen demand (BOD) levels of 900—6,000mg/l, usually ranging from 160 to 24,000mg/l – chemical oxygen demand (COD) ranging from 2,400 to 14,000mg/l – chloride ranging from 27,000mg/l and high levels of fat. Suspended solids are usually half of chloride levels.
- Wastewater may also contain residues of pesticides used to preserve hides during transport, as well as significant levels of pathogens.

- Significant volumes of solid wastes are produced, including trimmings, degraded hide, and hair from the beamhouse processes.
- The solid wastes can represent up to 70% of the wet weight of the original hides. In addition, large quantities of sludges are generated.
- Decaying organic material produces strong odours. Hydrogen sulfide is released during dehairing, and ammonia is released in deliming.
- Air quality may be further degraded by release of solvent vapours from spray application, decreasing, and finishing (for example, dye application).

Treatment Technologies

- Treatment of tannery wastewaters is always required. Some streams, such as slaking liquor (which has high salinity), sulfide-rich lime liquor, and chrome wastewaters should be segregated.
- Preliminary screening of wastewaters is required because of the large quantities of solids present.
- Recovery of hair from the dehairing and liming process reduces the BOD of the process effluent.
- Physical-chemical treatment precipitates metals and removes a large portion of solids, BOD, and COD.
- Biological treatment is usually required to reduce the remaining organic loads to acceptable levels (0.3kg BOD, 2kg COD, and 0.004kg chromium per metric ton of raw hide).
- Good ventilation and minimization of solvent release can avoid the need to collect and treat vapours in carbon adsorption beds, VOC emissions from finishing are approximately 30kg/t if pollution prevention measures are not adopted.
- Maximum upstream pollutant reduction is essential for tanneries, but treatment is also required.

3. Electroplating Industries

Introduction

Electroplating involves the deposition of a thin protective layer (usually metallic) onto a prepared metal surface, using electrochemical processes. The process involves pretreatment (cleaning, degreasing, and other preparation steps), plating, rinsing, passivating, and drying. The cleaning and pretreatment stages involve a variety of solvents (often chlorinated hydrocarbons, whose use is discouraged) and surface stripping agents, including caustic soda and a range of strong acids, depending on the metal surface to be plated.

In the plating process, the object to be plated is usually used as the cathode in an electrolytic bath. Plating solutions are acid or alkaline and may contain complexing agents such as cyanides.

Waste Characteristics

- Any or all of the substances used in electroplating (such as acidic solutions, toxic metals, solvents, and cyanides) can be found in the wastewater, either via rinsing of the product or from spillage and dumping of process baths.
- The solvents and vapours from hot plating baths result in elevated levels of volatile metal compounds, which may contain chromates.
- The mixing of cyanide and acidic wastewaters can generate lethal hydrogen cyanide gas, and this must be avoided. The overall wastewater stream is usually high in heavy metals, including cadmium, chrome, lead copper, zinc, and nickel, and in cyanides, fluorides, and oil and grease., all of which are process dependent
- Air emissions may contain toxic organics such as trichloroethylene and trichloroethane.
- Cleaning or changing of process tanks and treatment of wastewaters can generate substantial quantities of wet sludges containing high levels of toxic organics or metals.

Treatment Technologies

- Segregation of waste streams is essential because of the dangerous reactions that can occur. Strong acid and caustic reactions can generate boiling and splashing of corrosive liquids; acids can react with cyanides and generate lethal hydrogen cyanide gas. In addition, segregated streams that are concentrated are easier to treat.
- Exhaust hoods and good ventilation systems protect the working environment, but the exhaust streams should be treated to reduce VOCs and heavy metals to acceptable levels before venting to the atmosphere.
- Acid mists and vapors should be scrubbed with water before venting. In some cases, VOC levels of the vapors are reduced by use of carbon filters, which allow the reuse of solvents, or by combustion (and energy recovery) after scrubbing, adsorption, or other treatment methods.
- Cyanide destruction, flow equalization and neutralization, and metals removal are required, as a minimum, for electroplating plants.
- Individual design is necessary to address the characteristics of the specific plant, but there are a number of common treatment steps.
- Cyanide destruction must be carried out upstream of the other treatment processes.
- If hexavalent chrome (Cr^{+6}) occurs in the wastewaters, the wastewater is usually pretreated to reduce the chromium to a trivalent form using reducing agent, such as a sulfide.
- The main treatment processes are equalization, pH adjustment for precipitation, flocculation, and sedimentation/ filtration.
- The optimum pH for metal precipitation is usually in the range 8.5-11, but this depends on the mixture of metals present.

- Flocculation agents are sometimes used to facilitate the filtration of suspend solids.
- Modern wastewater treatment systems use ion exchange, membrane filtration, and evaporation to reduce the release of toxins and the quantity of effluent that needs to be discharged.
- Treatment sludges contain high levels of metals, and these should normally be managed as hazardous waste or sent for metals recovery.
- Electrolytical methods may be used to recover metals. Sludges are usually thickened, dewatered, and stabilized using chemical agents (such as lime) before disposal, which must be in an approved and controlled landfill.

4. Pulp and Paper Industries

Introduction

Pulp and paper are manufactured from raw materials containing cellulose fibers, generally wood, recycled paper, and agricultural residues.

The main steps in pulp and paper manufacturing are raw material preparation, such as wood debarking and chip making; paper manufacturing; and fiber recycling. Pulp mills and paper mills may exist separately or as integrated operations. Manufactured pulp is used as a source of cellulose for fiber manufacture and for conversion into paper or cardboard.

Chemical pulps are made by cooking (digesting) the raw materials, using the kraft (sulfate) and sulfite processes. Kraft processes produce a variety of pulps used mainly for packaging and high-strength papers and board.

Mechanical pulp can be used without bleaching to make printing papers for applications in which low brightness is acceptable-primarily, newsprint. However, for most printing, for copying, and some packaging grades, the pulp has to be bleached.

The soluble organic substances removed from the pulp in bleaching stages that use chlorine or chlorine compounds, as well as the substances removed in the subsequent alkaline stages, are chlorinated. Some of these chlorinated organic substances are toxic; they include dioxins, chlorinated phenols, and many other chemicals.

The finished pulp may be dried for shipment (market pulp) or may be used to manufacture paper on site.

Waste Characteristics

- The significant environmental impacts of the manufacture of pulp and paper result from the pulping and bleaching processes.
- In some processes, sulfur compounds and nitrogen oxides are emitted to the air, and chlorinated and organic compounds, nutrients, and metals are chlorinated to the wastewaters.

- In the pulping process, emissions of reduced sulfur compounds, measured as total reduced sulfur (TRS) and including hydrogen sulfide, methyl mercaptan, dimethyl sulfide, and dimethyl disulfide, are emitted, other pollutants are particulate matter, sulfur oxides, nitrogen oxides, and volatile organic compounds (VOCs),
- Wastewaters are discharged are high in biochemical oxygen demand (BOD), total suspended solids, chemical oxygen demand (COD), and chlorinated organic compounds, which may include dioxins, furans, and other adsorbable organic halides,
- Phosphorus and nitrogen are also released into wastewater. The main source of nutrients, nitrogen, and phosphorous compounds is raw material such as wood. The use of peroxide, ozone, and other chemicals in bleaching makes it necessary to use a complexing agent for heavy metals such as manganese.

Treatment Technologies

- Sulfur oxide emissions are scrubbed with slightly alkaline solutions. The reduced sulfur-compounds gases are collected using headers, hoods, and venting equipment. Condensates from the digester relief condenser and evaporation of black liquor are stripped of reduced sulfur compound. The stripper overhead and noncondensable are incinerated in a lime kiln or a dedicated combustion unit.
- Electrostatic precipitators are used to control the release of particulate matter into the atmosphere.
- Wastewater treatment typically includes (a) neutralization screening, sedimentation, and floatation/hydrocycloning to remove suspended solids and (b) biological/secondary treatment to reduce the organic content in wastewater and destroy toxic organics.
- Chemical precipitation is also used to remove certain cations. Fibers collected in primary treatment should be recovered and recycled.
- Biological treatment systems, such as activated sludge, aerated lagoons, and anaerobic fermentation, can reduce BOD by over 99%.
- Tertiary treatment may be performed to reduce toxicity, suspended solids, and color.
- Solid waste treatment steps include dewatering of sludge and combustion in an incinerator, bark boiler, or fossil-fuel-fired boiler.

5. Food Industries

- (i) **Dairy : Sources of Waste Waters :** Spillage, washing of cans, water from butter and cheese producing units.

Characteristics : High organic content, high BOD, Foul odour easily putrescible

Treatment : Equalization, Skimming, Grease removal, Biological treatment

(ii) **Distilleries :****Sources of waste waters :** Washings, Spillage**Characteristics :** Low pH, High organic matter, very high COD High suspended and dissolved organic matter.**Treatment :** Equalization, Biological treatment(iii) **Cane Sugar Industry :****Sources :** Cane wash water, effluents from evaporators, spray overflow, molasses effluents**Characteristics :** Coloured, disagreeable odour, contains oil and grease, high BOD, high carbonaceous matter**Treatment :** Biological treatment methods are not effective due to high concentration of carbohydrates.

Lagooning, oxidation ponds have been found to be the most practical treatment which is preceded by equalization and anaerobic digestion.

WEATHER

By absorbing infra-red (as well as by its contact with the hot ground), air heats up. As hot air expands, each cubic meter of it weighs less than before heating. Where the heating is most pronounced, the warm air is more buoyant than the cooler air surrounding it, and tends to float upwards; soaring birds and glider pilots seek such "thermal currents" and allow themselves to be carried upwards by them. This buoyancy is the basic process responsible for weather.

Rising air expands, and expansion of a gas cools it down, which is why mountain tops are cooler. Ultimately, a height is reached where not enough air remains on top to stop the IR radiation from escaping to space. The air then cools by radiation and stops rising, producing a relatively stable layer of the atmosphere known as the **stratosphere**. As one goes up in the stratosphere, air actually gets warmer from the absorption of ultra-violet light by ozone, and this extra heating keeps it even more stable.

Just below the boundary of the stratosphere ("tropopause"), air which has cooled is forced down again by warmer air rising from below. The result is a **circulation** of air, rising hot and returning cold, going around again and again, a motion known as **convection**. On a cold winter day such convection also occurs in homes: near poorly insulated windows the air cools and descends while further inside the room it rises again. The region between the ground and the stratosphere where convection and weather take place is known as the **troposphere**.

Sunlight also evaporates water from the oceans, from lakes and rivers and from green plants. Energy is invested in turning liquid water into vapour, and therefore humid air has **more** energy stored in it than dry air.

The capacity of air to hold water vapour depends strongly on temperature, and is smaller in cold air (just as less sugar can dissolve in cold water). As warm humid air rises, it expands and cools, and

since it then cannot hold as much water as before, the excess is forced out: Initially into the tiny droplets of clouds, then if the cooling is more drastic, into raindrops.

The remaining air is drier and **warmer** - warmed by water vapor turning back to liquid and returning energy to its surroundings - and warmer air is better able to radiate its heat into space. That is how water, clouds and rain play a major role in the transport of solar heat from the ground back to space and help create the complex patterns of weather and climate.

EXERCISES

Short Answer Questions

1. What are hydrosphere, lithosphere and biosphere?
2. What are primary and secondary pollutants?
3. What are biodegradable and non-biodegradable pollutions?
4. What is the compound formed when CO combines with blood?
5. How are NO and NO₂ formed in the atmosphere?
6. What are the reactions involved in removing SO₂ from the atmosphere by passing it through a solution containing citrate ions?
7. What type of aromatic compounds are present as particulates in the air?
8. What is the composition of pure dry air? What do you mean by the term 'sink' with respect to pollution.
9. What are the main air pollutants? Write about the sources and harmful effects of any one of them.
10. How carbon monoxide acts as a poison for human beings?
11. What remedial steps should be taken to save a person suffering from CO poisoning?
12. What are sources and sinks of nitrogen oxides as air pollutant? What are their effects?
13. How can pollution due to nitrogen oxides be controlled?
14. What are sources and sinks of sulphur oxide pollution.
15. What are particulates? What are their sources?
16. What are the harmful effects of particulate pollutants? List atleast five of them.
17. Write a short note on carbon dioxide sequestration.
18. List the main points of different between 'London smog' and 'Photochemical smog'.
19. What is 'Acid Rain'? What are harmful effects?
20. Brief explain 'Electrostatic precipitation' method for controlling particulate pollution.
21. What is 'Greenhouse effect'? How does it affect the global climate?
22. What are the factors responsible for depletion of ozone layer? What are the chemical reactions involved?
23. How synthetic detergents present as water pollutant create problems?
24. What are oxygen-demanding wastes? Define 'BOD₅'. How is it determined?

25. How plant nutrient and pesticides act as water pollutants?
26. What are thermal pollutants? What are their sources? What damage is done by them?

Long Answer Questions

1. What are the main components of our environment. Explain each of them briefly.
2. What is Air Pollution? What are the main sources of air pollution.
3. What are major air pollutants. Describe sources of pollution, sinks, harmful effects and methods to control the following pollutants: CO, NO_x, and SO_x.
4. What are the various chemical species and particulates present in the atmosphere?
5. Write informative notes on:
 - (a) Photochemical smog
 - (b) Green House Effect.
6. What are the major gases responsible for causing green house effect and how they are released into the atmosphere.
7. (a) What are the possible consequences of uncontrolled global warming.
(b) Write the chemical reactions involved in the depletion of ozone by nitric oxide.
8. List the auto-exhaust gases which contribute in polluting our environment. How could we control such pollution?
9. Mention the sources of particulate matter that pollute air. Explain the method for removal of particulate by Cottrell precipitator.
10. What is photochemical smog? Discuss the mechanism of photochemical smog formation with relevant reactions.
11. Mention the sources of water pollution.
12. Explain the various steps involved in primary treatment of sewage. Discuss the active sludge process.
13. Explain the terms BOD and COD in industrial waste water or distinguish between BOD and COD.
14. What is the difference between aerobic and anaerobic oxidation of sewage?
15. Write notes on: (i) sewage, (ii) primary and secondary air pollutants, (iii) trickling filter process, (iv) biological oxygen demand, (v) activated sludge process.
16. (a) What is BOD? Explain.
(b) Explain the primary treatment methods adopted for domestic sewage.
17. Name the principal sources of water pollution. Discuss briefly treatment of domestic sewage.
18. Write an explanatory note on "Acid Rain."
19. What is "Greenhouse effect"? Why is it so called? What are its consequences?
20. How can you classify water pollution into different types? Briefly explain each one of them.